Math

Class performance ৭ মার্ক থাকবে

Class ~~the~~ Test ৭ টা → 4 টার Best 3

① তৃতীয় সপ্তাহে একটা
4/5 তম সপ্তাহে ৫+২

formula class এর প্রথমে পড়াবে তারপর মডেল

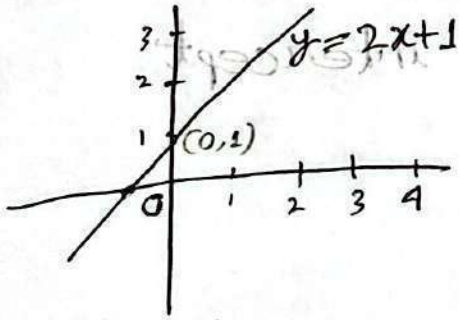
30 মার্ক আমার Class test + Presence + Performance থেকে

28 July 2026

Straight line

A straight line graph is a visual representation of a linear function. A straight line has a general equation of $y = mx + c$.

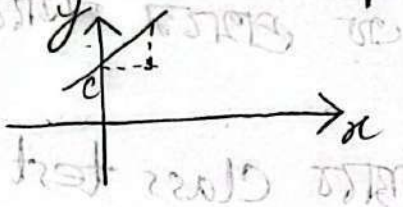
Ex:- For $y = 2x + 1$, $m = 2$, $c = 1$



Forms of straight line:-

① Slope - Intercept form:

$y = mx + c$, m is the slope and c is the y-intercept.



② Point Slope Form:

$y - y_1 = m(x - x_1)$, where m is the slope and (x_1, y_1) is a point on the line.

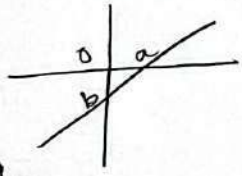
③ Standard form:

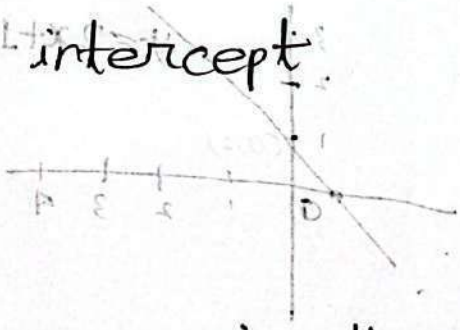
$ax + by = c$, where a, b, c are constants

$$\boxed{ax + by - c = 0}$$

④ Intercept form:

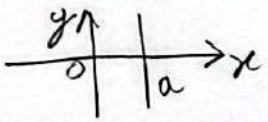
$\frac{x}{a} + \frac{y}{b} = 1$; where a is the x-intercept and b is the y-intercept.





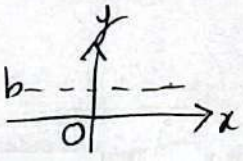
⑤ Vertical form:-

$x = a$, Vertical line with a is the x intercept



⑥ Horizontal form:

$y = b$. horizontal line with b as the y-intercept



$$m = \text{slope} = \frac{\text{difference in } y}{\text{difference in } x}$$

Two point slope form:

The equation of a straight line passing through two points (x_1, y_1) and (x_2, y_2)

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

Ex:

Find the equation of a line passing through the point $(1, -2)$ $(4, 5)$.

Soln:-

We know that two point slope form of straight line is

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y+2}{-2-5} = \frac{x-1}{1-4}$$

$$\Rightarrow \frac{y+2}{-7} = \frac{x-1}{-3}$$

$$\Rightarrow 3y + 6 = 7x - 7$$

$$\Rightarrow 7x - 3y = 13$$

(2,3), (5,7) বিন্দুগামী রেখার সমীকরণ

H.W. → ~~7x-8y=1~~ সমীকরণ

Parallel and Perpendicular line

(1) The equation of line parallel to $ax+by+c=0$ is $ax+by+k=0$

(2) The equation of line perpendicular to $ax+by+c=0$ is $bx-ay+k=0$

where k is an arbitrary constant to be determined by a suitable condition.

Ex: Find the equation of line passing through (-5,6) and parallel to $7x-8y=9$.

Sol^n:

The parallel line of $7x-8y=9$ is $7x-8y=k$ ... (1)

As the line is passing through (-5,6) then (1) becomes -

$$7(-5)-8\times 6=k$$

$$\Rightarrow -35-48=k$$

$$\therefore k=-83$$

Thus the required line is from (1)

$$7x-8y+83=0$$

W. H

Ex:-
Find the equation of line passing through (-5, 6) and perpendicular to $7x - 8y - 9 = 0$

Sol^n
We know that, the equation of line perpendicular to $ax + by + c = 0$ is $bx + ay + k = 0$
where k is an arbitrary constant

$a = 7, b = -8$ Thus the perpendicular line is

$$-8x - 7y + k = 0 \quad \text{--- (1)}$$

As the perpendicular line is passing through (-5, 6)

Thus, $-8(-5) - 7(6) + k = 0$

$\Rightarrow 40 - 42 + k = 0$

$\therefore k = 2$

(1) becomes,

$-8x - 7y + 2 = 0$

$\therefore 8x + 7y - 2 = 0$

H.W

H.W

Find the equation of lines passing through $(-1,2)$ and (a) parallal (b) Perpendicular to $3x-4y=6$.

Ans: (a) $3x-4y=-11$, (b) $-4x-3y=-2$

The equation of a line passing through the intersection of two lines:-

The equation of line passing through the two lines $ax+by+c_1=0$ and $a_1x+b_1y+c_1=0$ is

$$ax+by+c+\lambda(a_1x+b_1y+c_1)=0$$

where λ is an arbitrary constant.

Ex:- Find the equation of line passing through the intersection of two lines $2x+3y-5=0$, $x-y+2=0$

Soln: We know that,

The equation of line passing through the two lines $ax+by+c=0$ and $a_1x+b_1y+c_1=0$ is

$$ax+by+c+\lambda(a_1x+b_1y+c_1)=0$$

where λ is an arbitrary constant

So, $(2x+3y-5)+\lambda(x-y+2)=0$

$\Rightarrow 2x+3y-5+\lambda x-\lambda y+2\lambda=0$

$\Rightarrow (2+\lambda)x+(3-\lambda)y+(-5+2\lambda)=0$

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H.W

W.H

Similarly,

Ex:- $L_1 = 3x + y - 3 = 0$ (Ans) $(3+5\lambda)x + (1+\lambda)y + (3+6\lambda) = 0$

$L_2 = 5x + y + 6 = 0$

Sol^n: We know that,

The equation of line passing through the two lines $ax+by+c=0$ and $a_1x+b_1y+c_1=0$ is $ax+by+c+\lambda(a_1x+b_1y+c_1)=0$ where λ is an arbitrary constant

So, $3x+y-3+\lambda(5x+y-6)=0$

$\Rightarrow 3x+y-3+5\lambda x+\lambda y-6\lambda=0$

$\Rightarrow (3+5\lambda)x+(1+\lambda)y-(3+6\lambda)=0$

Perpendicular distance:

The length of the perpendicular distance from a point (x_1, y_1) on the line, $ax+by+c=0$ is

$$d = \pm \frac{ax_1+by_1+c}{\sqrt{a^2+b^2}}$$

Ex:-

Find the perpendicular distance from the point $(3,4)$ to the line $5x-12y+9=0$

Sol^n

Here $a=5$, $b=-12$, $c=9$

$$D = \frac{|5.3+(-12).4+9|}{\sqrt{5^2+(-12)^2}} \Rightarrow \frac{|15-48+9|}{\sqrt{25+144}} = \frac{24}{13}$$

H.W

Find the perpendicular distance from the point $(2,-3)$ to the line $3x+4y-10=0$ [Ans: $\frac{16}{5}$]

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H.W

1. Find the equation of a line passing through the point (2,3), (5,7)

$\Rightarrow$ We know that two point slope form of straight line is

$$\frac{y-y_1}{y_1-y_2} = \frac{x-x_1}{x_1-x_2}$$

$$\Rightarrow \frac{y-3}{3-7} = \frac{x-2}{2-5}$$

$$\Rightarrow \frac{y-3}{-4} = \frac{x-2}{-3}$$

$$\Rightarrow \frac{y-3}{4} = \frac{x-2}{3}$$

$$\Rightarrow 3y-9 = 4x-8$$

$$\Rightarrow 4x-3y+1 = 0$$

2(a). Find the equation of lines passing through (-1,2) and parallal to $3x-4y=6$

$\Rightarrow$ The parallal line of $3x-4y=6$ is $3x-4y=k$ ... (1)

As the line passing through (-1,2) the (1) becomes

$$3(-1)-4\times 2 = k$$

$$\Rightarrow -3-8 = k$$

$$\Rightarrow k = -11$$

Thus the requierd line from equation (1) $\rightarrow$

$$3x-4y = -11$$

H.W

2.(b) Find the equation of lines passing through $(-1,2)$ and perpendicular to $3x-4y=6$

$\Rightarrow$
We know that, the equation of line perpendicular to $ax+by+c=0$ is $bx-ay+k=0$
where k is an arbitrary constant
$a=3$, $b=-4$ Thus the perpendular line is

$$-4x-3xy+k=0 \quad \cdots\cdots (1)$$

As the perpendicular line is passing throug $(-1,2)$
Thus $-4(-1)-3\times 2+k=0$

$$\Rightarrow 4-6+k=0$$

$$\Rightarrow k=2$$

(1) becomes

$$-4x-3y+2=0$$

3. Find the perpendicular distance from the point $(2,-3)$ to the line $3x+4y-10=0$

$\Rightarrow$
Here, $a=3$, $b=4$, $c=-10$

$$\therefore D=\frac{|3\times 2+4\times(-3)-10|}{\sqrt{3^2+4^2}}=\frac{|6-12-10|}{5}$$

$$=\frac{16}{5}$$

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2/8/2026

~~Frequency Distribution~~

The condition of collinearity of three points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) is

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

* Check whether the points $(1,2), (3,6)$ and $(5,10)$ are collinear or not.

Sol^n :-

$$\begin{vmatrix} 1 & 2 & 1 \\ 3 & 6 & 1 \\ 5 & 10 & 1 \end{vmatrix}$$

$$= 1\begin{vmatrix} 6 & 1 \\ 10 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 1 \\ 5 & 1 \end{vmatrix} + 1\begin{vmatrix} 3 & 6 \\ 5 & 10 \end{vmatrix}$$

$$= (6-10) - 2(3-5) + 1(30-30)$$

$$= -4 - 2(-2) + 1 \cdot 0$$

$$= -4 + 4 + 0$$

$$= 0$$

Thus the points are collinear

Ex:- Check whether the points $(-1,1), (2,5), (3,6)$ are collinear or not.

Sol^n:-

$$\begin{vmatrix} -1 & 1 & 1 \\ 2 & 5 & 1 \\ 3 & 6 & 1 \end{vmatrix} = -1\begin{vmatrix} 5 & 1 \\ 6 & 1 \end{vmatrix} - 1\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix}$$

$$= -1(5-6) - 1(2-3) + 1(12-15)$$

$$= -(-1) - 1(-1) + 1(-3)$$

$$= 1 + 1 - 3$$

$$= -1$$

Thus the points are not collinear

Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represents a pair of parallel lines.

Sol^n:-

$$x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$$

$\Rightarrow x^2 + x(6y+4) + (9y^2 + 12y - 5) = 0$ [Quadratic in x]

which is in the form $ax^2 + bx + c = 0$

where $a = 1$,

$b = 6y + 4$

$c = 9y^2 + 12 - 5$

for $ax^2 + bx + c = 0$, roots are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\therefore x = \frac{-(6y+4) \pm \sqrt{(6y+4)^2 - 4.1.(9y^2+12y-5)}}{2.1}$$

$$= \frac{(-6y-4) \pm \sqrt{(6y)^2 + 2.6y.4 + (4)^2 - (36y^2+48y-20)}}{2}$$

$$x = \frac{-6y-4 \pm 6}{2}$$

$$x = -3y - 2 \pm 3$$

$$\therefore x + 3y + 5 = 0$$

$$x + 3y - 1 = 0$$

As these two lines differ only in constant terms, so these form a pair of paralled straight lines

Prove that the equation $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$ represents a pair of ~~parallel~~ perpendicular ~~lines~~ lines

Soln $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$

$\Rightarrow -3x^2 + x(-8y - 29) + (3y^2 + 3y - 18) = 0$

Comparing with the quadratic equation

$ax^2 + bx + c = 0$

here, $a = -3$

$b = -8y - 29$

and $c = 3y^2 + 3y - 18$

Thus the roots are,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-8y-29) \pm \sqrt{(-8y-29)^2 - 4(-3)(3y^2+3y-18)}}{2(-3)}$$

$$\therefore x = \frac{8y+29 \pm \sqrt{(-8y)^2 - 2(-8y).29 + (29)^2 + 12(3y^2+3y-18)}}{-6}$$

$$= \frac{8y+29 \pm \sqrt{64y^2+464y+841+36y^2+36y-216}}{-6}$$

$$\Rightarrow x = \frac{8y+29 \pm \sqrt{100y^2+500y+625}}{-6}$$

$$\Rightarrow x = \frac{8y+29 \pm \sqrt{(10y)^2 + 2.10y.25 + (25)^2}}{-6}$$

$$\Rightarrow x = \frac{8y+29 \pm \sqrt{(10y+25)^2}}{-6}$$

$$\Rightarrow x = \frac{8y+29 \pm (10y+25)}{-6}$$

Now,

$$x = \frac{8y+29+10y+25}{-6}$$

$$\Rightarrow x = \frac{18y+54}{-6}$$

$$\Rightarrow -6x = 18y+54$$

$$\boxed{6x+18y+54=0}$$

and

$$x = \frac{8y+29-10y-25}{-6}$$

$$\therefore x = \frac{-2y+4}{-6}$$

$$\therefore -6x = -2y+4$$

$$\boxed{6x-2y+4=0}$$

Here,

$$6x-2y+4=0 \quad \cdots (I)$$

$$6x+18y+54=0 \quad \cdots (II)$$

Formula

$ax+by+c$ এর ঢাল $\boxed{m=-\frac{a}{b}}$

Slope of ①, $m_1=\frac{-6}{-2}=3$, $m_2=-\frac{6}{18}=-\frac{1}{3}$

Since. $m_1 \cdot m_2 = 3\left(-\frac{1}{3}\right)=-1$

Thus the lines are perpendicular.

Class Work

1 * $2y^2-xy-x^2+y+2x-1=0$

H.W 2 * $x^2-5xy-6y^2+14x+5y+4=0$

H.W 3 * $2y^2+3xy-5y-6x+2=0$

} Next class এ জিজ্ঞাসা করবে

Sol(1) | C.W |

$2y^2-xy-x^2+y+2x-1=0$

$\Rightarrow -x^2+x(-y+2)+(2y^2+y-1)=0$

Comparing with the quadratic equation $ax^2+bx+c=0$

here, $a=-1$

$b=-y+2$

$c=2y^2+y-1$

Thus the roots are,

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$\Rightarrow x=\frac{-(-y+2)\pm\sqrt{(-y+2)^2-4(-1)(2y^2+y-1)}}{2(-1)}$$

$$\Rightarrow x = \frac{y-2 \pm \sqrt{y^2-4y+4+8y^2+4y-4}}{-2}$$

$$\Rightarrow x = \frac{y-2 \pm \sqrt{9y^2}}{-2}$$

$$\Rightarrow x = \frac{y-2 \pm 3y}{-2}$$

$$\Rightarrow -2x = y-2 \pm 3y$$

$$\Rightarrow 2x + y \pm 3y - 2 = 0$$

$$\therefore 2x + 4y - 2 = 0 \quad \ldots (i)$$

and $2x - 2y - 2 = 0 \quad \ldots (ii)$

~~Slope of (i), $m = \frac{-2}{4} \Rightarrow -\frac{1}{2}$, $m_2 = \frac{(-2)}{2} = 1$~~

Slope of (i), $m_1 = \frac{-2}{4} = -\frac{1}{2}$, $m_2 = \frac{-2}{-2} = 1$

Thus the lines are neither parallel nor perpendicular.

H.W

2. $x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$

$\Rightarrow x^2 + x(-5y + 14) + (-6y^2 + 5y + 4) = 0$

Comparing with the quadratic equation $ax^2 + bx + c = 0$

here, $a = 1$, $b = (-5y + 14)$ and $c = -6y^2 + 5y + 4$

Thus the roots are,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow x = \frac{-(-5y + 14) \pm \sqrt{(-5y + 14)^2 - 4 \cdot 1 \cdot (-6y^2 + 5y + 4)}}{2 \cdot 1}$$

$$\Rightarrow x = \frac{5y - 14 \pm \sqrt{25y^2 - 140y + 196 + 24y^2 - 20y \pm 16}}{2}$$

$$\Rightarrow 2x = 5y - 14 \pm \sqrt{49y^2 - 160y + 180}$$

$\Rightarrow$

~~Questions of Answer [illegible]~~

~~2. $x^2 - 5xy - 6y^2 + 14x + 5y + 34 = 0$~~

~~$\Rightarrow x^2 + x(-5y + 14) + (-6y^2 + 5y + 34) = 0$~~

~~[illegible]~~

H.W

3] $2y^2 + 3xy - 5y - 6x + 2 = 0$

$\Rightarrow 2y^2 + y(3x-5) + (-6x+2) = 0$

Comparing with the quadratic equation $ay^2+by+c=0$ here, $a=2$, $b=3x-5$ and $c=-6x+2$

Thus the roots are,

$$y = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$\Rightarrow y = \frac{-(3x-5) \pm \sqrt{(3x-5)^2 - 4\times 2\times(-6x+2)}}{2\times 2}$$

$$\Rightarrow 4y = -3x+5 \pm \sqrt{9x^2-30x+25+48x-16}$$

$$\Rightarrow 4y = -3x+5 \pm \sqrt{9x^2+18x+9}$$

$$\Rightarrow 4y = -3x+5 \pm \sqrt{(3x)^2+2\cdot 3x\cdot 3+3^2}$$

$$\Rightarrow 4y = -3x+5 \pm \sqrt{(3x+3)^2}$$

$$\Rightarrow 4y = -3x+5 \pm (3x+3)$$

Now,

$4y = -3x+5+(3x+3)$

$\Rightarrow 4y = 8$

$\Rightarrow 4y - 8 = 0$

$\Rightarrow y - 2 = 0 \quad \cdots (1)$

And,

$4y = -3x+5-(3x+3)$

$\Rightarrow 4y = -3x+5-3x-3$

$\Rightarrow 4y + 6x - 2 = 0$

$\Rightarrow 3x + 2y - 1 = 0 \quad \cdots (2)$

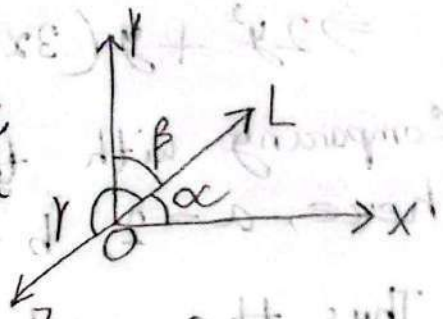
Slope of (1), $m_1 = -\frac{0}{1} = 0$, $m_2 = -\frac{3}{2}$

Thus the lines are neither parallel nor perpendicular.

04 August 2026

Direction Cosine of a line

If a given straight line OL makes angles α, β, γ with x-axis, y-axis, z-axis respectively then $\cos\alpha, \cos\beta, \cos\gamma$ are known as the direction cosine of the straight line OL which is denoted by l, m, n respectively.



That is $l = \cos\alpha,\ m = \cos\beta,\ n = \cos\gamma$

So, the direction cosine of

$$OL = (l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$$

Relation between Direction Cosines:-

The sum of squares of direction cosines of every line is one. That is if l, m, n be the direction cosines of any line, then

$$l^2 + m^2 + n^2 = 1$$

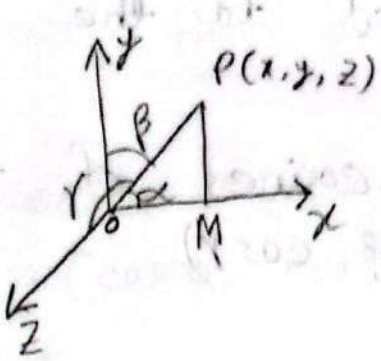
Proof:-

Let the direction cosines of the line OP be $(l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$ as the coordinates of P be (x, y, z)

Here OP is the position vector,

$$OP = \sqrt{x^2 + y^2 + z^2}$$

$\therefore OP^2 = x^2 + y^2 + z^2 \quad \ldots\ldots (i)$



Draw a perpendicular line PM from P on the x-axis then $OM = x$

From $\triangle OPM$, $\cos\alpha = \frac{OM}{OP} = \frac{x}{OP}$

$\therefore x = OP\cos\alpha$

$\Rightarrow x = l\,OP \quad \ldots\ldots (ii)$

Similarly, $y = m \cdot OP \quad \ldots (iii)$

$z = n \cdot OP \quad \ldots (iv)$

Now, $(ii)^2 + (iii)^2 + (iv)^2$, we get

$x^2 + y^2 + z^2 = l^2 OP^2 + m^2 OP^2 + n^2 OP^2$

$\Rightarrow OP^2 = OP^2(l^2 + m^2 + n^2)$ [Using (i)]

$\Rightarrow l^2 + m^2 + n^2 = 1$

Or, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Define direction cosines and find the relationship between them.

Ex:- Find the direction cosine of the line which is equally inclined to the axes

Soln:-

Let the direction cosines of the line be

$(l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$

Here,

$\alpha = \beta = \gamma$

$\therefore \cos\alpha = \cos\beta = \cos\gamma$

Thus we can write $l = m = n$ ... (i)

We know that,

$l^2 + m^2 + n^2 = 1$

$\Rightarrow l^2 + l^2 + l^2 = 1$ [using (i)]

$\Rightarrow 3l^2 = 1$

$\therefore l^2 = \frac{1}{3}$

$\therefore l = \pm\frac{1}{\sqrt{3}}$

Similarly,

$m = \pm\frac{1}{\sqrt{3}}$

$n = \pm\frac{1}{\sqrt{3}}$

So the direction cosines are $(l, m, n) = \left(\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right)$

Ex:- If a line makes angles α, β, γ with the axes, then show that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Sol^n:-

Let the direction cosines of the line be

$(l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$

We know that,

$$l^2 + m^2 + n^2 = 1$$

$$\Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\Rightarrow (1 - \sin^2\alpha) + (1 - \sin^2\beta) + (1 - \sin^2\gamma) = 1$$

$$\Rightarrow 3 - (\sin^2\alpha + \sin^2\beta + \sin^2\gamma) = 1$$

$$\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$$

Direction Ratio

Any three numbers a, b, c which are proportional to the direction cosines l, m, n respectively of a given line are called the direction ratios of the given line.

By definition $a \propto l$

$$\therefore a = kl \quad \ldots (i)$$

k is a constant

$b \propto m$

$$b = km \quad \ldots (ii)$$

Define direction ratio
Relationship between direction cosines and direction ratio

$c \propto n$

$c = kn \quad \ldots (iii)$

Now, $(i)^2 + (ii)^2 + (iii)^2$, we get,

$$a^2+b^2+c^2 = l^2k^2 + m^2k^2 + n^2k^2$$

$$\Rightarrow a^2+b^2+c^2 = k^2(l^2+m^2+n^2)$$

$$\Rightarrow a^2+b^2+c^2 = k^2 \cdot 1$$

$$\Rightarrow k^2 = a^2+b^2+c^2$$

$$\Rightarrow k = \pm\sqrt{a^2+b^2+c^2}$$

Putting the values of k in (i), (ii), (iii) we get,

$$a = \pm\sqrt{a^2+b^2+c^2} \times l$$

$$\therefore l = \pm \frac{a}{\sqrt{a^2+b^2+c^2}}$$

Similarly, $m = \pm \frac{b}{\sqrt{a^2+b^2+c^2}}$

and $n = \pm \frac{c}{\sqrt{a^2+b^2+c^2}}$

are the mathematical representation of the relationship between direction cosines and direction ratio of a line.

9 August 2026

Formula 1:-

If a, b, c are direction ratios of the line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$

then, $a = x_2 - x_1$, $b = y_2 - y_1$, $c = z_2 - z_1$

Problem:-

If P and Q are $(2, 3, -6)$ and $(3, -4, 5)$ respectively and O be the origin, then find the direction cosines of OP, OQ, PQ.

Soln:-

The coordinate of origin O is $(0, 0, 0)$

Direction Cosines of OP:

Let a_1, b_1, c_1 be the direction ratios of OP and l_1, m_1, n_1 be the direction cosines of OP.

Now, $a_1 = 2 - 0 = 2$

$b_1 = 3 - 0 = 3$

$c_1 = -6 - 0 = -6$

So direction cosines of OP:

$$l_1 = \frac{a_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{2}{\sqrt{2^2 + 3^2 + (-6)^2}} = \frac{2}{\sqrt{49}} = \frac{2}{7}$$

$$m_1 = \frac{b_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{3}{\sqrt{2^2 + 3^2 + (-6)^2}} = \frac{3}{\sqrt{49}} = \frac{3}{7}$$

$$n_1 = \frac{c_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{-6}{\sqrt{2^2 + 3^2 + (-6)^2}} = \frac{-6}{\sqrt{49}} = -\frac{6}{7}$$

OQ = ?

PQ = ?



$l_2 = \frac{3}{5\sqrt{2}}$ $m_2 = \frac{-4}{5\sqrt{2}}$ $n_2 = \frac{1}{\sqrt{2}}$

$l_3 = \frac{1}{\sqrt{171}}$ $m_3 = \frac{-7}{\sqrt{171}}$ $n_3 = \frac{11}{\sqrt{171}}$

Direction Cosines of OQ

Let a_2, b_2, c_2 be the direction ratios of OQ and l_2, m_2, n_2 be the direction cosines of OQ

Now, $a_2 = 3-0 = 3$

$b_2 = -4-0 = -4$

$c_2 = 5-0 = 5$

So direction cosines of OQ:

$$l_2 = \frac{a_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{3}{\sqrt{3^2+(-4)^2+5^2}} = \frac{3}{\sqrt{50}} = \frac{3}{5\sqrt{2}}$$

$$m_2 = \frac{b_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{-4}{\sqrt{3^2+(-4)^2+5^2}} = \frac{-4}{\sqrt{50}} = \frac{-4}{5\sqrt{2}}$$

$$n_2 = \frac{c_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{5}{\sqrt{3^2+(-4)^2+5^2}} = \frac{5}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Direction cosines of PQ

Let a_3, b_3, c_3 be the direction ratios of PQ and l_3, m_3, n_3 be the direction cosines of PQ

Now, $a_3 = 3-2 = 1$

$b_3 = -4-3 = -7$

$c_3 = 5+6 = 11$



So direction cosines of PQ

$$l_3 = \frac{a_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{1}{\sqrt{1^2+(-7)^2+(11)^2}} = \frac{1}{\sqrt{171}}$$

$$m_3 = \frac{b_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{-7}{\sqrt{1^2+(-7)^2+(11)^2}} = \frac{-7}{\sqrt{171}}$$

$$n_3 = \frac{c_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{11}{\sqrt{1^2+(-7)^2+(11)^2}} = \frac{11}{\sqrt{171}}$$

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H.W: If P and Q are (2, 3, 4) and (1, 1, −1) respectively and O be the origin, then find the direction cosines of OP, OQ and PQ

ans: $OP\left(\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}}\right)$ $OQ\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$ $PQ\left(-\frac{1}{\sqrt{30}}, \frac{-2}{\sqrt{30}}, \frac{-5}{\sqrt{30}}\right)$

Formula 2:-

If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines and θ be the angle between them, then

$$\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

Formula 3:-

Condition of perpendicularity of two lines:
If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines and if these two lines are perpendicular, then

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

Formula 4:-

Condition of Parallelism of two lines:
If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines and if these two lines are parallel, then

$$l_1 = l_2, \quad m_1 = m_2, \quad n_1 = n_2$$



Problem:-

Find the cosine of the angle between two lines having the direction cosines $\left(\frac{4}{9}, -\frac{8}{9}, \frac{-1}{9}\right)$ and $\left(-\frac{1}{3}, \frac{-2}{3}, -\frac{2}{3}\right)$.

Soln:-

$l_1 = \frac{4}{9}$,

$m_1 = \frac{-8}{9}$,

$n_1 = -\frac{1}{9}$

$l_2 = -\frac{1}{3}$;

$m_2 = \frac{-2}{3}$,

$n_2 = \frac{-2}{3}$



Let θ be the angle between two lines,

then $\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$

$= \left(\frac{4}{9}\right)\cdot\left(-\frac{1}{3}\right) + \left(-\frac{8}{9}\right)\cdot\left(-\frac{2}{3}\right) + \left(-\frac{1}{9}\right)\cdot\left(\frac{-2}{3}\right)$

$= -\frac{4}{27} + \frac{16}{27} + \frac{2}{27}$

$\therefore \cos\theta = \frac{14}{27}$

H.W → Find the cosine of the angle between two lines having the direction cosines $\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and $\left(\frac{1}{2}, \frac{-\sqrt{6}}{\sqrt{4}}, -\frac{1}{2}\right)$

Ans: $\cos\theta = \frac{1}{\sqrt{2}}$

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H.W → Check that joining points from A(5, 2, -3) to Q(6, 1, 4) to line joining the points from C(-3, 2, -1) to L(-1, -4, 1) is parallel or not

Problem:-

Prove that, joining the points from P(2, 3, -2) to Q(3, 1, 1) is parallel to the line joining the points from R(2, 1, -5) to L(4, -3, 1)

Sol^n:-

Direction cosines of PQ:

Let a_1, b_1, c_1 be the direction ratios of PQ and l_1, m_1, n_1 be the direction cosines of PQ

Now,

$a_1 = 3-2 = 1$, $b_1 = 1-3 = -2$ $c_1 = 1-(-2) = 3$

So the direction cosine of PQ:

$$l_1 = \frac{1}{\sqrt{1^2+(-2)^2+3^2}} = \frac{1}{\sqrt{14}}$$

$$m_1 = \frac{-2}{\sqrt{1^2+(-2)^2+3^2}} = \frac{-2}{\sqrt{14}}$$

$$n_1 = \frac{3}{\sqrt{1^2+(-2)^2+3^2}} = \frac{3}{\sqrt{14}}$$



Direction cosines of RL:

Let, a_2, b_2, c_2 be the direction ratios of RL and l_2, m_2, n_2 be the direction cosines of RL

Now, $a_2 = 4-2 = 2$, $b_2 = -3-1 = -4$, $c_2 = 1-(-5) = 6$

So direction cosines of RL

$$l_2 = \frac{2}{\sqrt{2^2+(-4)^2+6^2}} = \frac{2}{\sqrt{56}} = \frac{2}{2\sqrt{14}} = \frac{1}{\sqrt{14}}$$

$$m_2 = \frac{-4}{\sqrt{2^2+(-4)^2+6^2}} = \frac{-4}{\sqrt{56}} = \frac{-4}{2\sqrt{14}} = \frac{-2}{\sqrt{14}}$$

$$n_2 = \frac{6}{\sqrt{2^2+(-4)^2+6^2}} = \frac{6}{\sqrt{56}} = \frac{6}{2\sqrt{14}} = \frac{3}{\sqrt{14}}$$

Since, $l_1 = l_2$, $m_1 = m_2$ and $n_1 = n_2$, so PQ and RL lines are parallel

Problem: Prove that, the lines AB and BC are perpendicular to each other when A(-11, 8, 4) B(-1, -7, -1), C(9, -2, 4)

Soln:

D.C of AB:

Let a_1, b_1, c_1 be the direction ratios of AB and l_1, m_1, n_1 be the direction cosines of AB

$a_1 = -1 - (-11) = 10$

$b_1 = -7 - 8 = -15$

$c_1 = -1 - 4 = -5$

So direction cosines of AB:

$$l_1 = \frac{10}{\sqrt{10^2+(-15)^2+(-5)^2}} = \frac{10}{\sqrt{350}}$$

$$m_1 = \frac{-15}{\sqrt{10^2+(-15)^2+(-5)^2}} = \frac{-15}{\sqrt{350}}$$

$$n_1 = \frac{-5}{\sqrt{10^2+(-15)^2+(-5)^2}} = \frac{-5}{\sqrt{350}}$$



D.C. of BC:-

Let a_2, b_2, c_2 be the direction ratios of BC and l_2, m_2, n_2 be the direction cosines of BC

Now,

$a_2 = 9 - (-1) = 10$,

$b_2 = -2 - (-7) = 5$

$c_2 = 4 - (-1) = 5$

So DC of BC:

$$l_2 = \frac{10}{\sqrt{10^2+5^2+5^2}} = \frac{10}{\sqrt{150}}$$

$$m_2 = \frac{5}{\sqrt{10^2+5^2+5^2}} = \frac{5}{\sqrt{150}}$$

$$n_2 = \frac{5}{\sqrt{10^2+5^2+5^2}} = \frac{5}{\sqrt{150}}$$

Now, $l_1l_2 + m_1m_2 + n_1n_2$

$= \frac{10}{\sqrt{350}} \times \frac{10}{\sqrt{150}} + \left(\frac{-15}{\sqrt{350}} \times \frac{5}{\sqrt{150}}\right) + \left(\frac{-5}{\sqrt{350}} \times \frac{5}{\sqrt{150}}\right)$

$= 0$ ∴ The lines are perpendicular

H:W

Prove that the lines AB and CD are perpendicular to each other where.

A(6, -6, 0),

B(-1, -7, 6), C(3, -4, 4)

D(2, -9, 2)

H.W.

1) If P and Q are (2, 3, 4) and (1, 1, −1) respectively and O be the origin, then find the direction Cosines of OP, OQ and PQ.

Sol^n:

The coordinate of origin O is (0, 0, 0)

Direction cosines of OP:

Let a_1, b_1, c_1 be the direction ratios of OP and l_1, m_1, n_1 be the direction cosines of OP

Now,

$a_1 = 2-0 = 2$, $b_1 = 3-0 = 3$, $c_1 = 4-0 = 4$

So direction cosines of OP:

$$l_1 = \frac{a_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{2}{\sqrt{2^2+3^2+4^2}} = \frac{2}{\sqrt{29}}$$

$$m_1 = \frac{a_2}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{3}{\sqrt{2^2+3^2+4^2}} = \frac{3}{\sqrt{29}}$$

$$n_1 = \frac{a_3}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{4}{\sqrt{2^2+3^2+4^2}} = \frac{4}{\sqrt{29}}$$

Direction cosines of OQ:

Let a_2, b_2, c_2 be the direction ratios of OQ and l_2, m_2, n_2 be the direction cosines of OQ

Now, $a_2 = 1-0 = 1$, $b_2 = 1-0 = 1$, $c_2 = -1-0 = -1$

So direction cosines of OQ:

$$l_2 = \frac{a_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{1}{\sqrt{1^2+1^2+(-1)^2}} = \frac{1}{\sqrt{3}}$$

$$m_2 = \frac{b_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{1}{\sqrt{1^2+1^2+(-1)^2}} = \frac{1}{\sqrt{3}}$$

$$n_2 = \frac{c_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{-1}{\sqrt{1^2+1^2+(-1)^2}} = \frac{-1}{\sqrt{3}}$$



H.W

Direction cosines of PQ:

Let a_3, b_3, c_3 be the direction ratios of PQ and l_3, m_3, n_3 be the direction cosines of PQ

Now,

$a_3 = 1-2 = -1$, $b_3 = 1-3 = -2$, $c_3 = -1-4 = -5$

So direction cosines of PQ:

$$l_3 = \frac{a_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{-1}{\sqrt{(-1)^2+(-2)^2+(-5)^2}} = \frac{-1}{\sqrt{1+4+25}} = \frac{-1}{\sqrt{30}}$$

$$m_3 = \frac{b_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{-2}{\sqrt{(-1)^2+(-2)^2+(-5)^2}} = \frac{-2}{\sqrt{1+4+25}} = \frac{-2}{\sqrt{30}}$$

$$n_3 = \frac{c_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{-5}{\sqrt{(-1)^2+(-2)^2+(-5)^2}} = \frac{-5}{\sqrt{1+4+25}} = \frac{-5}{\sqrt{30}}$$

2| Find the cosine of the angle between two lines having the direction cosines $\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and $\left(\frac{1}{2}, \sqrt{\frac{6}{4}}, -\frac{1}{2}\right)$

Soln:

$l_1 = \frac{1}{\sqrt{3}}$, $m_1 = \frac{-1}{\sqrt{3}}$, $n_1 = \frac{1}{\sqrt{3}}$

$l_2 = \frac{1}{2}$, $m_2 = \sqrt{\frac{6}{4}}$, $n_2 = \frac{-1}{2}$

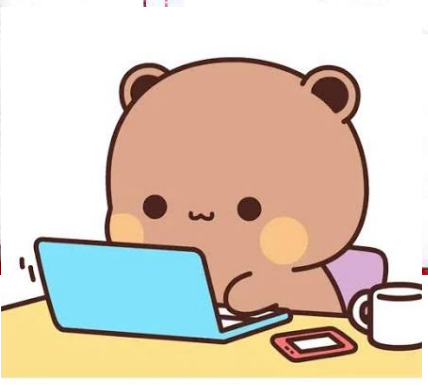
Let θ be the angle between two lines, then, $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$

$$= \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{3}}\right) + \left(-\frac{1}{\sqrt{3}}\right)\left(\sqrt{\frac{6}{4}}\right) + \left(\frac{1}{\sqrt{3}}\right)\left(\frac{-1}{2}\right)$$

$$= \frac{1}{2\sqrt{3}} + \frac{\sqrt{6}}{2\sqrt{3}} - \frac{1}{2\sqrt{3}}$$

$$= \frac{1+\sqrt{6}-1}{2\sqrt{3}}$$

$$= \frac{1}{\sqrt{2}}$$



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H.W.



3] Check that joining points from A(5,2,-3) to Q(6,1,4) to line joining the points from C(-3,2,-1) to L(-1,-4,13) is parallal or not

Soln:-

Direction cosines of AQ:

Let a_1, b_1, c_1 be the direction ratios of AQ and l_1, m_1, n_1 be the direction cosines of AQ

now, $a_1 = 6-5 = 1$, $b_1 = 1-2 = -1$, $c_1 = 4+3 = 7$

So, the direction cosines of AQ

$$l_1 = \frac{a_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{1}{\sqrt{(1)^2+(-1)^2+7^2}} = \frac{1}{\sqrt{1+1+49}} = \frac{1}{\sqrt{51}}$$

$$m_1 = \frac{b_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{-1}{\sqrt{(1)^2+(-1)^2+7^2}} = \frac{-1}{\sqrt{1+1+49}} = \frac{-1}{\sqrt{51}}$$

$$n_1 = \frac{c_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{7}{\sqrt{(1)^2+(-1)^2+(7)^2}} = \frac{7}{\sqrt{1+1+49}} = \frac{7}{\sqrt{51}}$$

Direction cosines of CL:

Let a_2, b_2, c_2 be the direction ratios of CL and l_2, m_2, n_2 be the direction cosines of CL

now, $a_2 = -1+3 = 2$, $b_2 = -4-2 = -6$, $c_2 = 13+1 = 14$

So, the direction cosines of CL

$$l_2 = \frac{a_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{2}{\sqrt{2^2+(-6)^2+(14)^2}} = \frac{2}{\sqrt{236}} = \frac{2}{2\sqrt{59}} = \frac{1}{\sqrt{59}}$$

$$m_2 = \frac{b_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{-6}{\sqrt{2^2+(-6)^2+(14)^2}} = \frac{-6}{2\sqrt{59}} = \frac{-3}{\sqrt{59}}$$

$$n_2 = \frac{c_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{14}{2\sqrt{59}} = \frac{7}{\sqrt{59}}$$

Since, $l_1 \neq l_2$, $m_1 \neq m_2$, $n_1 \neq n_2$

So, AQ and CL lines are not parallal

H.W

4] Prove that the lines AB and CD are perpendicular to each other where A(6,-6,0), B(-1,-7,6), C(3,-4,4), D(2,-9,2)

Sol^n:-

Direction cosines of AB:

Let a_1, b_1, c_1 be the direction ratios of AB and l_1, m_1, n_1 be the direction cosines of AB

$a_1 = -1-6 = -7$, $b_1 = -7+6 = -1$, $c_1 = 6-0 = 6$

So direction cosines of AB:

$$l_1 = \frac{a_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{-7}{\sqrt{(-7)^2+(-1)^2+6^2}} = \frac{-7}{\sqrt{49+1+36}} = \frac{-7}{\sqrt{86}}$$

$$m_1 = \frac{a_2}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{-1}{\sqrt{(-7)^2+(-1)^2+6^2}} = \frac{-1}{\sqrt{49+1+36}} = \frac{-1}{\sqrt{86}}$$

$$n_1 = \frac{a_3}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{6}{\sqrt{(-7)^2+(-1)^2+6^2}} = \frac{6}{\sqrt{49+1+36}} = \frac{6}{\sqrt{86}}$$

Direction cosines of CD:

Let a_2, b_2, c_2 be the direction ratios of CD and l_2, m_2, n_2 be the direction cosines of CD

$a_2 = 2-3 = -1$, $b_2 = -9+4 = -5$, $c_2 = 2-4 = -2$

So direction cosines of CD:

$$l_2 = \frac{a_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{-1}{\sqrt{(-1)^2+(-5)^2+(-2)^2}} = \frac{-1}{\sqrt{1+25+4}} = \frac{-1}{\sqrt{30}}$$

$$m_2 = \frac{-5}{\sqrt{30}}$$

$$n_2 = \frac{-2}{\sqrt{30}}$$

Now, $l_1 l_2 + m_1 m_2 + n_1 n_2$

$$= \left(\frac{-7}{\sqrt{86}}\right)\left(\frac{-1}{\sqrt{30}}\right) + \left(\frac{-1}{\sqrt{86}}\right)\left(\frac{-5}{\sqrt{30}}\right) + \left(\frac{6}{\sqrt{86}}\right)\left(\frac{-2}{\sqrt{30}}\right)$$

$$= 0$$

∴ The lines are perpendicular

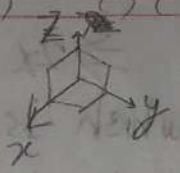


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18/8/2026

Plane



General Equation of Plane:

$ax+by+cz+d=0$; where a, b, c, d are constants.

Formula 1: General equation of plane passing through the origin $(0,0,0)$ is $ax+by+cz=0$

Formula 2: Equation of plane passing through the point (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) is

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

Formula 3: The perpendicular distance from the point (x_1, y_1, z_1) to the plane $ax+by+cz=0$ is

$$D = \left| \frac{ax_1+by_1+cz_1+d}{\sqrt{a^2+b^2+c^2}} \right|$$

Find equation of plane passing through $(2,3,1)$, $(1,1,3)$, $(2,2,3)$. Also find perpendicular distance from the point $(5,6,7)$

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 2 & 2 & 3 & 1 \end{vmatrix} = \begin{vmatrix} x-2 & y-3 & z-1 & 0 \\ 1 & 2 & -2 & 0 \\ -1 & -1 & 0 & 0 \\ 2 & 2 & 3 & 1 \end{vmatrix} \quad \begin{bmatrix} R_1' = R_1 - R_2 \\ R_2' = R_2 - R_3 \\ R_3' = R_3 - R_4 \end{bmatrix}$$

$$= -1\{(y-3)(-2) - 2(z-1)\} + 1\{(x-2)(-2) - (z-1)\}$$

$$= 2y - 6 + 2z - 2 - 2x + 4 - z + 1 = -2x + 2y + z - 3 = 0$$

$\Rightarrow 2x-2y-z+3=0$

which is the required equation of a plane

The perpendicular distance from the point (5,6,7) to this plane is

~~Practice: Fin~~ $\left|\frac{2\times5-2\times6-1\times7+3}{\sqrt{2^2+(-2)^2+(-1)^2}}\right|$

$$=\left|\frac{10-12-7+3}{\sqrt{4+4+1}}\right|=\left|\frac{-6}{\sqrt{9}}\right|=|-2|=2$$

Practiee:

Find the equation of plane which is passing through (2,1,3), (-1,-2,4), (4,2,1). Also find the perpendicular distance from the point (1, 1, -1) to this plane.

$\Rightarrow$ The equation of plane passing through (2, 1, 3), (-1,-2, 4), (4, 2, 1) is

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 1 & 3 & 1 \\ -1 & -2 & 4 & 1 \\ 4 & 2 & 1 & 1 \end{vmatrix} = 0$$

here,

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 1 & 3 & 1 \\ -1 & -2 & 4 & 1 \\ 4 & 2 & 1 & 1 \end{vmatrix} = \begin{vmatrix} x & y & z & 1 \\ 2-x & 1-y & 3-z & 0 \\ -3 & -3 & 1 & 0 \\ 5 & 4 & -3 & 0 \end{vmatrix}$$

$[R_4' = R_4 - R_3$
$R_3' = R_3 - R_2$
$R_2' = R_2 - R_1]$

$$= 1\begin{vmatrix} 2-x & 1-y & 3-z \\ -3 & -3 & 1 \\ 5 & 4 & -3 \end{vmatrix}$$

$$= 5\{(1-y)\times 1 - (-3)(3-z)\} - 4\{(2-x)\times 1 - (-3)(3-z)\} - 3\{(2-x)(-3) - (-3)(1-y)\}$$

$$= 5(1-y+9-3z) - 4(2-x+9-3z) - 3(-6+3x+3-3y)$$

$$= 5-5y+45-15z-8+4x-36+12z+18-9x-9+9y$$

$$= -5x+4y-3z+15=0$$

∴ Required equation

$$-5x+4y-3z+15=0$$

$$\Rightarrow 5x-4y+3z-15=0$$

which is the required equation of a plane

The perpendicular distance from the point $(1,1,-1)$ to this plane is

$$\left|\frac{5\times 1-4\times 1+3\times(-1)-15}{\sqrt{5^2+(-4)^2+3^2}}\right|$$

$$= \left|\frac{5-4-3-15}{\sqrt{50}}\right|$$

$$= \left|\frac{-17}{\sqrt{50}}\right|$$

$$= \frac{17}{\sqrt{50}}$$

Formula 4:

The four point $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3), (x_4, y_4, z_4)$ are coplanar if their determinant is zero.

Problem:
Show that the four points $(0, -1, -1), (4, 5, 1), (3, 9, 4), (-4, 4, 4)$ are coplanar.

Solution:

To prove that the four points are coplanar, we need to show their determinant is zero,

Now,
$$\begin{vmatrix} 0 & -1 & -1 & 1 \\ 4 & 5 & 1 & 1 \\ 3 & 9 & 4 & 1 \\ -4 & 4 & 4 & 1 \end{vmatrix}$$

$$= 0\begin{vmatrix} 5 & 1 & 1 \\ 9 & 4 & 1 \\ 4 & 4 & 1 \end{vmatrix} + 1\begin{vmatrix} 4 & -1 & 1 \\ 3 & 4 & 1 \\ -4 & 4 & 1 \end{vmatrix} - 1\begin{vmatrix} 4 & 5 & 1 \\ 3 & 9 & 1 \\ -4 & 4 & 1 \end{vmatrix} - 1\begin{vmatrix} 4 & 5 & 1 \\ 3 & 9 & 4 \\ -4 & 4 & 4 \end{vmatrix}$$

$$= 0 + 1\{4(4-4) - 1(3+4) + 1(12+16)\} - 1\{4(9-4) - 5(3+4) + 1(12+36)\} - 1\{4(36-16) - 5(12+16) + 1(12+36)\}$$

$$= (-7+28) - (20 - 35 + 48) - (80 - 140 + 48)$$

$$= -7 + 28 - 20 + 35 - 48 - 80 + 140 - 48 = 0$$

So these four points are coplanar.

[Proved]

Practice

Show that the four points $(-6, 3, 2)$, $(3, -2, 4)$, $(5, 7, 3)$, $(-13, 17, -1)$ are coplanar

Solution

To prove that the four points are coplanar, we need to show their determinant is zero,

Now, $\begin{vmatrix} -6 & 3 & 2 & 1 \\ 3 & -2 & 4 & 1 \\ 5 & 7 & 3 & 1 \\ -13 & 17 & -1 & 1 \end{vmatrix}$

$= -6\begin{vmatrix} -2 & 4 & 1 \\ 7 & 3 & 1 \\ 17 & -1 & 1 \end{vmatrix} - 3\begin{vmatrix} 3 & 4 & 1 \\ 5 & 3 & 1 \\ -13 & -1 & 1 \end{vmatrix} + 2\begin{vmatrix} 3 & -2 & 1 \\ 5 & 7 & 1 \\ -13 & 17 & 1 \end{vmatrix} - 1\begin{vmatrix} 3 & -2 & 4 \\ 5 & 7 & 3 \\ -13 & 17 & -1 \end{vmatrix}$

$= -6\{(-2)(3+1) - 4(7-17) + 1(-7-51)\}$
$\quad -3\{3(3+1) - 4(5+13) + 1(-5+39)\}$
$\quad +2\{3(7-17) + 2(5+13) + 1(35+91)\}$
$\quad -1\{3(-7-51) + 2(-5+39) + 4(35+91)\}$

$= -6\{-8 + 40 - 58\} - 3\{12 - 72 + 34\} + 2\{-30 + 36 + 176\} - 1\{-174 + 68 + 704\}$

~~$= 48 - 240 + 348 - 36 + 216 - 102 + 472 + 172 + 174 - 68 - 704$~~
~~-60~~ ~~352~~

$= -6(-26) - 3(-26) + 2 \times 182 - 1 \times 598$

$= 0$

So the four points are coplanar

[Proved]

23/08/2026

Formula 5

Angle Between two planes:

If $a_1x + b_1y + c_1z + d_1 = 0$
and $a_2x + b_2y + c_2z + d_2 = 0$
are two planes, and θ be the angle between them,

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$



Problem:

Find the angle between two planes $2x - 2y - z + 3 = 0$ and $5x - 4y + 3z - 15 = 0$.

Soln:-

Let θ be the angle between these two planes,

$$\cos\theta = \frac{2\cdot5 + (-2)\times(-4) + (-1)\times3}{\sqrt{\{2^2+(-2)^2+(-1)^2\}\{5^2+(-4)^2+3^2\}}}$$

$$\Rightarrow \cos\theta = \frac{15}{\sqrt{9\times50}} = \frac{15}{3\times5\sqrt{2}}$$

$$\Rightarrow \cos\theta = \frac{1}{\sqrt{2}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \cos^{-1}\left(\cos\frac{\pi}{4}\right)$$

$$\therefore \theta = \frac{\pi}{4}$$

H.W

Find the angle between two planes

$2x - y + z - 6 = 0$
and $x + y + 2z - 7 = 0$

Ans: $\theta = \frac{\pi}{3}$

Formula 6:-

Condition of perpendicularity of two planes:

If $a_1x + b_1y + c_1z + d_1 = 0$ and
$a_2x + b_2y + c_2z + d_2 = 0$ are two ~~perpendi~~ planes, and they are perpendicular to each other, then

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

Formula 7:

Condition of paralelism of two planes:

If $a_1x + b_1y + c_1z + d_1 = 0$ and
$a_2x + b_2y + c_2z + d_2 = 0$ are two plane, and they are paralled to each other, then

$$a_1 = a_2 ; \; b_1 = b_2 ; \; c_1 = c_2$$

Formula 8:

The equation of plane parallel to a given plane $ax + by + cz + d = 0$ is $ax + by + cz + k = 0$



Problem:-

Find the equation of plane which is passing through $(1,2,2)$ and parallel to $3x+2y+z-7=0$.

Soln:-

The equation of plane parallel to $3x+2y+z-7=0$ is $3x+2y+z+k=0 \ldots\ldots (1)$

As, (1) is passing through $(1,2,2)$, thus

$$(3\times1)+(2\times2)+2+k=0$$
$$\Rightarrow 9+k=0$$
$$\therefore k=-9$$

Putting the value of k in (1) we get,

$$3x+2y+z-9=0$$

(*) Find the equation of plane which is passing through $(4,0,1)$ and parallel to $4x+3y-12z+6=0$

Soln:-

The equation of plane parallel to $4x+3y-12z+6=0$ is $4x+3y-12z+k=0 \ldots (1)$

As (1) is passing through $(4,0,1)$, thus

$$4\times4+3\times0-12\times1+k=0$$
$$\Rightarrow 4+k=0$$
$$\therefore k=-4$$

Putting the value of k in (1) we get,

$$4x+3y-12z-4=0$$

Ans: $x+3y+z-2=0$

H.W.1 Find the equation plane which is passing through $(1,1,2)$ and perpendicular to the planes $2x-2y-4z-6=0$ and $3x+y+6z-4=0$

Ans

Formula 9:-

Equation of plane passing through the point (x_1, y_1, z_1) is $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$

Problem:-

Find the equation of plane which is passing through $(-1,3,2)$ and perpendicular to two planes $x+2y+2z-5=0$ and $3x+3y+2z-8=0$

Soln:-

The equation of plane passing through $(-1,3,2)$ is $a(x+1)+b(y-3)+c(z-2)=0 \ldots\ldots (I)$

Since, (I) is perpendicular to two planes

$x+2y+2z-5=0$ and $3x+3y+2z-8=0$

So we get,

$1.a+2.b+2c=0 \ldots\ldots (II)$

$3.a+3b+2c=0 \ldots\ldots (III)$

By cross multiplication is (II), (III) we get,

$$\frac{a}{4-6}=\frac{-b}{2-6}=\frac{c}{3-6}$$

$$\therefore \frac{a}{-2}=\frac{b}{4}=\frac{c}{-3}=k \text{ (say)}$$

So, $a=-2k$, $b=4k$, $c=-3k$

Putting the values of a, b, c in (I) we get,

$-2k(x+1)+4k(y-3)-3k(z-2)=0$

$\Rightarrow -2(x+1)+4(y-3)-3(z-2)=0$

$\Rightarrow -2x-2+4y-12-3z+6=0$

$\Rightarrow -2x+4y-3z-8=0$

Which is the required plane.

H.W

Find the equation of plane which is passing through $(1,-1,1)$ and the intersection of two planes $x-2y+3z+4=0$ and $2x-3y+4z-7=0$ [Ans: $9x-5y+17z-39=0$]

Formula 10:-

The equation of plane passing through the interaction of two planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ is

$$(a_1x+b_1y+c_1z+d_1)+K(a_2x+b_2y+c_2z+d_2)=0$$

Problem:-

Find the equation of plane which is passing through $(1,2,3)$ and the intersection of two planes $x+2y+3z+4=0$ and $4x+3y+2z+1=0$

Solution:-

The equation of plane passing through the intersection of two planes is

$$(x+2y+3y+4)+K(4x+3y+2z+1)=0 \quad \cdots\cdots (1)$$

we get,

$$1+(2\times 2)+3\times 3+4+K\{(4\times 1)+(3\times 2)+(2\times 3)+1\}=0$$

$$\Rightarrow (1+4+9+4)+K(4+6+6+1)=0$$

$$\Rightarrow 18+17K=0$$

$$\therefore K=-\frac{18}{17}$$

Putting the value of K in (1) we get,

$$(x+2y+3z+4)-\frac{18}{17}(4x+3y+2z+1)=0$$

$$\Rightarrow 17(x+2y+3z+4)-18(4x+3y+2z+1)=0$$

$$\Rightarrow 17x+34y+51z+68-72x-54y-36z-18=0$$

$$\Rightarrow -11x-4y+3z+10=0$$

which is the required plane

Problem:

Find the equation of plane which is parallel to $4x - 4y + 7z - 3 = 0$ and a distance 4 unit from point $(4, 1, -2)$

Sol^n:-

The equation of plane parallel to $4x - 4y + 7z - 3 = 0$ is

$$4x - 4y + 7z + K = 0 \quad \cdots \textcircled{1}$$

The distance from point $(4, 1, -2)$ to ① is

$$\pm \frac{4\times4 - 4\times1 + 7\times(-2) + K}{\sqrt{4^2 + (-4)^2 + (7)^2}}$$

Thus $\pm\left(\frac{-2+K}{\sqrt{81}}\right) = 4$

$\Rightarrow \pm(-2+K) = 36$

Now,

$-2 + K = 36$

$\therefore K = 38$

$-(-2+K) = 36$

$\Rightarrow 2 - K = 36$

$\therefore K = -34$



Putting the value of K in ① we get

$$\begin{cases} 4x - 4y + 7z + 38 = 0 \\ 4x - 4y + 7z - 34 = 0 \end{cases}$$

H.W

Find the equation of plane which is parallel to plane $2x - 3y - 6z - 14 = 0$ and a distance 5 unit from the origin.

Ans: $\begin{cases} 2x - 3y - 6z + 35 = 0 \\ 2x - 3y - 6z - 35 = 0 \end{cases}$

CamScanner

H.W. Starts here

1. Find the angle between two planes $2x-y+z-6=0$ and $x+y+2z-7=0$

Ans:

Let θ be the angle between these two planes,

$$\cos\theta = \frac{2\times1+(-1)\times1+(1)\times2}{\sqrt{2^2+(-1)^2+(1)^2}\sqrt{1^2+1^2+2^2}}$$

$$\Rightarrow \cos\theta = \frac{2-1+2}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

2. Find the equation plane which is passing through $(1,1,2)$ and perpendicular to the plane $2x-2y-4z-6=0$ and $3x+y+6z-4=0$

Ans:

The equation of plane passing through $(1,1,2)$ is

$$a(x-1)+b(y-1)+c(z-2)=0 \quad \ldots (1)$$

Since (1) is perpendicular to two planes $2x-2y-4z-6=0$ and $3x+y+6z-4=0$

So we get,

$$2a-2b-4c=0 \quad \ldots (2)$$

$$3a+b+6c=0 \quad \ldots (3)$$

By cross multiplication in (2) and (3) we get-

$$\frac{a}{-12+4} = \frac{b}{-12-12} = \frac{c}{2+6}$$

$$\Rightarrow \frac{a}{-8} = \frac{b}{-24} = \frac{c}{8}$$

$$\Rightarrow \frac{a}{-1} = \frac{b}{-3} = c = K \text{ (say)}$$

So, $a=-K$, $b=3K$, $c=K$

Cross multiplication

$a_1x+b_1y+c_1z=0$
$a_2x+b_2y+c_2z=0$

$a_1 \; b_1 \; c_1 \; a_1$
$a_2 \; b_2 \; c_2 \; a_2$

$\Rightarrow \frac{x}{b_1c_2-b_2c_1} = \frac{y}{c_1a_2-c_2a_1} = \frac{z}{a_1b_2-a_2b_1}$

Putting the values of a, b, c in (1) we get,

$-1K(x-1)-3K(y-1)+1K(z-2)=0$

$\Rightarrow -1(x-1)-3(y-1)+1(z-2)=0$

$\Rightarrow -1x+1-3y+3+z-2=0$

$\Rightarrow x+3y-z-2=0$

3. Find the equation of plane which is passing through $(1,-1,1)$ and the intersection of two planes $x-2y+3z+4=0$ and $2x-3y+4z-7=0$

Ans:

The equation of plane passing through the intersection of two planes is

$$x-2y+3z+4+K(2x-3y+4z-7)=0 \quad \ldots (1)$$

Since (1) is passing through $(1,-1,1)$, we we get,

$1-2\times(-1)+3\times1+4+K(2\times1-3\times(-1)+4\times1-7)=0$

$\Rightarrow 1+2+3+4+K(2+3+4-7)=0$

$\Rightarrow 10+K(2)=0$

$\Rightarrow K=-\frac{10}{2}=-5$

Putting value of K in (1) we get

$x-2y+3z+4-5(2x-3y+4z-7)=0$

$\Rightarrow x-2y+3z+4-10x+15y-20z+35=0$

$\Rightarrow -9x+13y-17z+39=0$

$\Rightarrow 9x-13y+17z-39=0$ which is the required plane

A.W ends here

4. Find the equation of plane which is parallel to plane $2x-3y-6z-14=0$ and a distance 5 unit from the origin

Ans:

The equation of plane parallel $2x-3y-6z-14=0$ is $2x-3y-6z+K=0$ —(1)

The distance from origin $(0,0,0)$ is ~~5 unit~~

$$\pm \frac{2\times0-3\times0-6\times0+K}{\sqrt{2^2+(-3)^2+(-6)^2}}$$



Thus $\pm \frac{K}{\sqrt{4+9+36}} = 5$

$\Rightarrow \pm \frac{K}{\sqrt{49}} = 5$

$\Rightarrow \pm \frac{K}{7} = 5$

$\Rightarrow K = \pm 35$

$\therefore K = 35$ and $K = -35$

Putting the value of K in (1) we get

$2x-3y-6z+35=0$

$2x-3y-6z-35=0$

25.08.2026

Defination exam এ আসবে পাবে

upper-triangular matrix

Echelon matrix (row echelon form)

① All zero rows appear at the bottom.

② Each non-zero row has a leading entry (Pivot), and each pivot is strictly to the right of the pivot in the row above.

③ All entries below each pivot are zero

$$\begin{pmatrix} 3 & 2 & 0 & 7 \\ 0 & 4 & 5 & 10 \\ 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 3 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

Reduced row echelon matrix

* উপরের ①, ②, ③ Copy-Paste

④ Each pivot equals 1.

⑤ Each pivot is the only non-zero entry in its column

$$\begin{pmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

* ● → Any scaler $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

☆ Define Echelon matrix
☆ Define row echelon matrix

Mid এ আসবে ☆☆☆☆☆

দুটি Matrix সমান হলে সবগুলোতে সমান এলিমেন্ট থাকতে হবে।
নিচে তাই (=) Use না করে (~) করা হয়েছে।

Ex:-
Find the Echelon form of $\begin{pmatrix} 3 & 4 & -1 & 2 \\ 1 & -2 & 3 & 1 \\ 3 & 14 & -11 & 1 \end{pmatrix}$

Sol^n:
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 3 & 4 & -1 & 2 \\ 3 & 14 & 11 & 1 \end{pmatrix} [R_1 \longleftrightarrow R_2]$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 10 & -10 & -1 \\ 0 & 20 & -20 & -2 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 10 & -10 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} [R_3' = R_3 - 2R_2]$$

Ex:-
Find the echelon form of $\begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 4 \\ 0 & 7 & 10 \end{pmatrix}$

Sol^n
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 10 \\ 0 & 7 & 10 \end{pmatrix} [R_2' = R_2 + 2R_1]$$

$$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 10 \\ 0 & 0 & 0 \end{pmatrix} [R_3' = R_3 - R_2]$$

Ex :- Find the echelon form of $\begin{pmatrix} 1 & -1 & 2 & 0 \\ 1 & 1 & 1 & 1 \\ -1 & -3 & 0 & -2 \end{pmatrix}$

Sol^n

Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & -4 & 2 & 2 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 + R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{bmatrix} R_3' = R_3 + 2R_2 \end{bmatrix}$$

H.W:

(1) $\begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$ (2) $\begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & 4 \end{pmatrix}$

(3) $\begin{pmatrix} 2 & 1 & -1 & 2 \\ -1 & 3 & 2 & 3 \\ 1 & 2 & 1 & -1 \\ 2 & -3 & -1 & 4 \end{pmatrix}$

AD BREAK

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H.W.

① Find the echelon form of $\begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$

Sol^n
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - R_2 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} R_3' = R_3 + R_2 \end{bmatrix}$$

② Find the echelon form of $\begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & 4 \end{pmatrix}$

Sol^n
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & 1 & 3 \\ 0 & 2 & -6 \\ 0 & 2 & -2 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 + 2 \times R_2 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 1 & 3 \\ 0 & 2 & -6 \\ 0 & 0 & -8 \end{pmatrix} \begin{bmatrix} R_3' = R_3 - R_2 \end{bmatrix}$$

③ Find the echelon form of $\begin{pmatrix} 2 & 1 & -1 & 2 \\ -1 & 3 & 2 & 3 \\ 1 & 2 & 1 & -1 \\ 2 & -3 & -1 & 4 \end{pmatrix}$

Sol^n
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 2 & 1 & -1 & 2 \\ 0 & 1 & 3 & 8 \\ 0 & 5 & 3 & 2 \\ 0 & -7 & -3 & 6 \end{pmatrix} \begin{bmatrix} R_4' = R_4 + 2R_3 \\ R_3' = R_3 + R_2 \\ R_2' = 2R_2 + R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 2 & 1 & -1 & 2 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & -12 & -38 \\ 0 & 0 & 6 & 44 \end{pmatrix} \begin{bmatrix} R_4' = 5R_4 + 7R_3 \\ R_3' = R_3 - 5R_2 \end{bmatrix}$$

$$\sim \begin{pmatrix} 2 & 1 & -1 & 2 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & -12 & -38 \\ 0 & 0 & 6 & 50 \end{pmatrix} \begin{bmatrix} R_4' = 2 \times R_4 + R_3 \end{bmatrix}$$

30 August 2026

Find the reduced row echelon form of $\begin{pmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{pmatrix}$

Soln:-

Reduce the matrix to reduced row echelon form by means of elementary operation.

$$\sim \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{pmatrix} \left[R_2' = \frac{R_2}{2}\right]$$

$$\sim \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{pmatrix} \left[R_3' = R_3 - 3R_2\right]$$

$$\sim \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{pmatrix} \left[R_3' = R_3(-2)\right]$$

$$\sim \begin{pmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{pmatrix} \left[R_1' = (-1)R_2 + R_1\right]$$

$$\sim \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{pmatrix} \begin{bmatrix} R_1' = \frac{-11}{2}R_3 + R_1 \\ R_2' = \frac{7}{2}R_3 + R_2 \end{bmatrix}$$

Which is the reduced row echelon matrix

(*) Find the reduced row echelon form of

$$\begin{pmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & 5 & 6 & -5 & -1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{pmatrix} \quad [R_1 \leftrightarrow R_2]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{pmatrix} \quad \left[R_1' = \frac{R_1}{2}\right]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{pmatrix} \quad [R_3' = R_3 - 2R_1]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{pmatrix} \quad \left[R_2' = R_2\left(-\frac{1}{2}\right)\right]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 1 \end{pmatrix} \quad [R_3' = R_3 - 5R_2]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad [R_3' = R_3 \cdot 2]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad \left[R_2' = \frac{7}{2}R_3 + R_2\right]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad [R_1' = -6R_3 + R_1]$$

$$\sim \begin{pmatrix} 1 & 2 & 0 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad [R_1' = 5R_2 + R_1]$$

CT ৪০ ইউনিট প্রশ্ন
একটা Algebric Plane আর একটা Matrix থেকে

* Find the reduced row echelon form of

$$\begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & -2 & 0 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{pmatrix} \quad [R_2' = R_2 - 2R_1,\ R_4' = R_4 - 2R_1]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{pmatrix} \quad [R_2' = (-1)R_2]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 \end{pmatrix} \quad [R_3' = R_3 - 5R_2,\ R_4' = R_4 - 4R_2]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_4' = R_4 \times \tfrac{1}{6}]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_3 \longleftrightarrow R_4]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_2' = 0 - 3R_3 + R_2]$$

$$\sim \begin{pmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_1' = 2R_2 + R_1]$$

1/9/26

Find the reduced row echelon form of

① $\begin{pmatrix} 3 & 4 & -1 & 2 \\ 1 & -2 & 3 & 1 \\ 3 & 14 & -11 & 1 \end{pmatrix}$ ② $\begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 4 \\ 0 & 7 & 10 \end{pmatrix}$

①
$$\begin{pmatrix} 3 & 4 & -1 & 2 \\ 1 & -2 & 3 & 1 \\ 3 & 14 & -11 & 1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 3 & 4 & -1 & 2 \\ 3 & 14 & -11 & 1 \end{pmatrix} \quad [R_1 \longleftrightarrow R_2]$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 10 & -10 & -1 \\ 0 & 20 & -20 & -2 \end{pmatrix} \quad \begin{bmatrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -1 & -\frac{1}{10} \\ 0 & 20 & -20 & -2 \end{pmatrix} \quad \left[R_2' = \frac{R_2}{10}\right]$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -1 & -\frac{1}{10} \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_3' = R_3 - 20R_2]$$

$$\sim \begin{pmatrix} 1 & 0 & 1 & \frac{4}{5} \\ 0 & 1 & -1 & -\frac{1}{10} \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_1' = R_1 + 2R_2]$$

(2) $\begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 1 \\ 0 & 7 & 10 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 10 \\ 0 & 7 & 10 \end{pmatrix}$ $[R_2' = R_2 + 2R_1]$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & \frac{10}{7} \\ 0 & 7 & 10 \end{pmatrix}$ $\left[R_2' = \frac{R_2}{7}\right]$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & \frac{10}{7} \\ 0 & 0 & 0 \end{pmatrix}$ $[R_3' = R_3 - 7R_2]$

$\sim \begin{pmatrix} 1 & 0 & \frac{1}{7} \\ 0 & 1 & \frac{10}{7} \\ 0 & 0 & 0 \end{pmatrix}$ $[R_1' = R_1 - 2R_2]$

(3) $\begin{pmatrix} 1 & 2 & -3 & -2 & -3 \\ 3 & 8 & -2 & 0 & -4 \\ 1 & 3 & -7 & -2 & -11 \\ 2 & 1 & -9 & -10 & -3 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 2 & -3 & -2 & -3 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 2 & 2 & 4 & -2 \\ 0 & -3 & -3 & -6 & 3 \end{pmatrix}$ $\begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - 3R_1 \\ R_4' = R_4 - 2R_1 \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 2 & -3 & -2 & -3 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ $\begin{bmatrix} R_3' = R_3 - 2R_2 \\ R_4' = R_4 + 3R_2 \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 0 & -5 & -6 & -1 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ $[R_1' = R_1 - 2R_2]$

8/9/2026

~~Using Row operations~~ Finding inverse matrix

For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 3 & 0 & 8 \end{pmatrix}$, the computations are as follows:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 3 & 0 & 8 & 0 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right) \begin{bmatrix} R_3 = R_3 + 2R_2 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \begin{bmatrix} R_3' = (-1) R_3 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \begin{bmatrix} R_1' = (-3) R_3 + R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \begin{bmatrix} R_1' = (-2) R_2 + R_1 \end{bmatrix}$$

Thus

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

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1. Find the inverse of the following matrix by elementary operations.

$$\begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$$

Sol^n:-

$$(A|I) = \left(\begin{array}{ccc|ccc} 3 & -3 & 4 & 1 & 0 & 0 \\ 2 & -3 & 4 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0 \\ 2 & -3 & 4/3 & -2/3 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \left[R_1' = \frac{R_1}{3}\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 4/3 & 1/3 & 0 & 0 \\ 0 & -1 & 4/3 & -2/3 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \left[R_2' = R_2 - 2R_1\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 4/3 & 1/3 & 0 & 0 \\ 0 & 1 & -4/3 & 2/3 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \left[R_2' = -R_2\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 4/3 & 1/3 & 0 & 0 \\ 0 & 1 & -4/3 & 2/3 & -1 & 0 \\ 0 & 0 & -1/3 & 2/3 & -1 & 1 \end{array}\right) \left[R_3' = R_3 + R_2\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 4/3 & 1/3 & 0 & 0 \\ 0 & 1 & -4/3 & 2/3 & -1 & 0 \\ 0 & 0 & 1 & -2 & 3 & -3 \end{array}\right) \left[R_3' = -3R_3\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 3 & -4 & 4 \\ 0 & 1 & 0 & -2 & 3 & -4 \\ 0 & 0 & 1 & -2 & 3 & -3 \end{array}\right) \left[\begin{array}{l} R_1' = -\frac{4}{3}R_3 + R_1 \\ R_2' = \frac{4}{3}R_3 + R_2 \end{array}\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -2 & 3 & -4 \\ 0 & 0 & 1 & -2 & 3 & -3 \end{array}\right) \left[R_1' = R_2 + R_1\right]$$

$$\sim (I \mid A^{-1})$$

Thus $A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{pmatrix}$

10/

Spring 2026

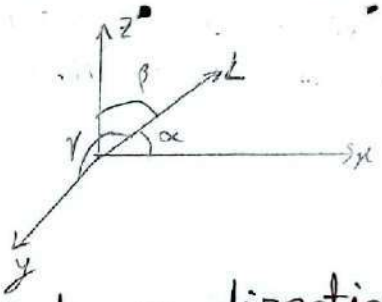
Class Test 1(Section A)	Program: B.Sc in CSE
Registration No.: 25101003	Course Title: Math III
Name: Utsob Ghosh	Course Code: Math 201
Time:30 minutes	Total Marks:10

1. Define Direction Cosine of a line. Find the relationship between Direction Cosines.
2. Find the perpendicular distance from the point (4, -3) to the line $5x + 3y - 11 = 0$.

Answer to the question no -1

If a line OL makes angle α, β, γ with axes x-axes, y-axes, z-axes respectively then $\cos\alpha$, $\cos\beta$, $\cos\gamma$ are called direction cosines. They are denoted as l, m, n respectively.

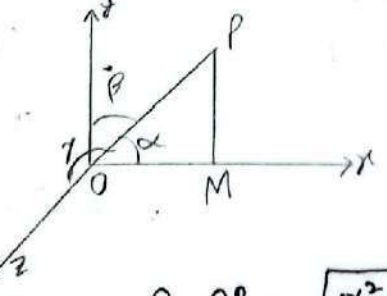
So, direction cosines of OL $= (l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$



Relationship between direction cosines

Let, a line OP make angle α, β, γ with x-axes, y-axes, z axes respectively. Direction cosines are l, m, n.

$l = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$



Position vector of $OP = \sqrt{x^2+y^2+z^2}$

$\Rightarrow OP^2 = x^2+y^2+z^2 \ldots\ldots (1)$

Draw a perpendicular PM on x-axes from P.

From $\triangle OPM$,

$$\cos\alpha = \frac{PM}{OP} = \frac{x}{OP}$$

$\Rightarrow x = OP\cos\alpha$

$\Rightarrow x = OP \times l$

Similarly, $y = OP \times m$ and $z = OP \times n$

$\therefore x^2+y^2+z^2 = (OP\times l)^2+(OP\times m)^2+(OP\times n)^2 = OP^2l^2+OP^2m^2+OP^2n^2$

$\Rightarrow x^2+y^2+z^2 = OP^2(l^2+m^2+n^2)$

$\Rightarrow OP^2 = OP^2(l^2+m^2+n^2)$ [From equation ①]

$\Rightarrow \frac{OP^2}{OP^2} = l^2+m^2+n^2$

$\Rightarrow 1 = l^2+m^2+n^2$

$\therefore l^2+m^2+n^2 = 1$

It is the relationship between direction cosines.

Answer to the question no-2

We know, If a line is $ax+by+c$ then the perpendicular line of the line is $bx-ay+k=0$

Given,

$5x+3y-11=0$

here, $a=5, b=3, c=-11$

∴ perpendicular line is $3x-5y+k=0 \ldots$ ①

Perpendicular distance from the point $(4,-3)$ to the line $5x+3y-11=0$ is $\frac{5\times4+3\times(-3)-11}{\sqrt{5^2+3^2}}$

$= \frac{20+(-9)-11}{\sqrt{25+9}}$

$= \frac{0}{\sqrt{34}}$

$= 0$

Since the point $(4,-3)$ is located in the line $5x+3y-11=0$ the perpendicular distance is 0.

Spring 2026

Class Test 2(Section A)	Program: B.Sc in CSE
Registration No.: 25101003	Course Title: Math III
Name: Utsob Ghosh	Course Code: Math 201
Time:20 minutes	Total Marks:10

1. Show that the four points (-6, 3, 2), (3, -2, 4), (5, 7, 3), (-13, 17, -1) are coplanar.
2. Find the reduced row Echelon form of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$.

1. The four points (-6, 3, 2), (3, -2, 4), (5, 7, 3), (-13, 17, -1) are co-planar if their determinant is zero.

Now,

$$\begin{vmatrix} -6 & 3 & 2 & 1 \\ 3 & -2 & 4 & 1 \\ 5 & 7 & 3 & 1 \\ -13 & 17 & -1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -6 & 3 & 2 & 1 \\ 9 & -5 & 2 & 0 \\ 11 & 4 & 1 & 0 \\ -7 & 14 & -3 & 0 \end{vmatrix} \quad \begin{bmatrix} R_4' = R_4 - R_1 \\ R_3' = R_3 - R_1 \\ R_2' = R_2 - R_1 \end{bmatrix}$$

$$= -1 \times \begin{vmatrix} 9 & -5 & 2 \\ 11 & 4 & 1 \\ -7 & 14 & -3 \end{vmatrix} + 0 - 0 + 0$$

$$= -1 \times \begin{vmatrix} 9 & 0 & 0 \\ 11 & 91 & -13 \\ -7 & 91 & -13 \end{vmatrix} \quad \begin{bmatrix} C_2' = 9C_2 + 5C_1 \\ C_3' = 9C_3 - 2C_1 \end{bmatrix}$$

$$= -1 \times 9 \times \{91 \times (-13) - 91 \times (-13)\} = -9(-1183 + 1183)$$

$$= -9 \times 0$$

$$= 0$$

Since their determinant is zero the four points are coplanar

2. $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & -2 & 5 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{pmatrix} \begin{bmatrix} R_3' = R_3 + 2R_2 \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 0 & 9 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{pmatrix} \begin{bmatrix} R_1' = R_1 - 2R_2 \end{bmatrix}$

~~which is the required reduced row echelon form of the matrix~~

$\sim \begin{pmatrix} 1 & 0 & 9 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} R_3' = \frac{R_3}{-1} \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} R_2' = R_2 + 3R_3 \\ R_1' = R_1 - 9R_3 \end{bmatrix}$

Which is the ~~requed~~ required reduced row echelon ~~matrix~~ form of the matrix.

27/8/2026 next class এ আগে নিবে

final এ নিবে ⇐ Final solution

Inversion Algorithm

A - Coefficient matrix

$$A\underline{x} = \underline{b}$$

$$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}(\underline{b})$$

$$\Rightarrow (A^{-1}A)\underline{x} = A^{-1}\underline{b}$$

$$\Rightarrow I\underline{x} = A^{-1}\underline{b}$$

$$\Rightarrow \underline{x} = A^{-1}\underline{b}$$

2. Solve the system by finding A^{-1} by using elementary operations

$$0 + 2x_2 + x_3 + 3x_4 = 1$$
$$x_1 + x_2 - x_3 - 2x_4 = -1$$
$$x_1 + 2x_2 + 0 + x_4 = 2$$
$$-x_1 + x_2 + 2x_3 + 6x_4 = 1$$

Where $A = \begin{pmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{pmatrix}$, $\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}$

$$\therefore (A \mid I) = \left(\begin{array}{cccc|cccc} 0 & 2 & 1 & 3 & 1 & 0 & 0 & 0 \\ 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 2 & 6 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 3 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 2 & 6 & 0 & 0 & 0 & 1 \end{array}\right) \quad [R_1 \leftrightarrow R_2]$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 3 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 2 & 1 & 4 & 0 & 1 & 0 & 1 \end{array}\right) \begin{array}{l} [R_3' = R_3 - R_1 \\ R_4' = R_4 + R_1] \end{array}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 2 & 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 4 & 0 & 1 & 0 & 1 \end{array}\right) [R_2 \longleftrightarrow R_3]$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -3 & 1 & 2 & -2 & 0 \\ 0 & 0 & -1 & -2 & 0 & 3 & -2 & 1 \end{array}\right) [R_3' = R_3 - 2R_2 \\ R_4' = R_4 - 2R_3]$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -3 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) [R_4' = R_4 - R_3]$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & -1 & -2 & 2 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) [R_3' = -R_3]$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & -1 & 0 & -2 & 3 & 0 & 2 \\ 0 & 1 & 1 & 0 & 3 & -4 & 1 & -3 \\ 0 & 0 & 1 & 0 & 2 & -5 & 2 & -3 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) \begin{array}{l} [R_1' = R_1 + 2R_4 \\ R_2' = R_2 - 3R_4 \\ R_3' = R_3 - 3R_4] \end{array}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 0 & -2 & 2 & -1 \\ 0 & 1 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 2 & -5 & 2 & -3 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) [R_1' = R_1 + R_3 \\ R_2' = R_2 - R_3]$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & -3 & 3 & -1 \\ 0 & 1 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 2 & -5 & 2 & -3 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 1 \end{array}\right) [R_1' = R_1 - R_2]$$

$\sim (I \mid A^{-1})$

$$A^{-1} = \begin{pmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{pmatrix}$$

$\rightarrow A\underline{x} = \underline{b}$

$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}\underline{b}$

$\Rightarrow \underline{x} = A^{-1}\underline{b}$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -1+3+6-1 \\ 1-1-2 \\ 2+5+4-3 \\ -1-1+0+1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 7 \\ -2 \\ 8 \\ -1 \end{pmatrix}$$

$3x-3y+4z=1$

$2x-3y+4z=2$

$0-y+z=1$

Here, $A=\begin{pmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & 1 & 1\end{pmatrix}$, $\underline{X}=\begin{pmatrix}x\\ y\\ z\end{pmatrix}$, $\underline{b}=\begin{pmatrix}1\\ 2\\ 1\end{pmatrix}$

$\therefore (A:I)$

$$=\left(\begin{array}{ccc|ccc}3 & -3 & 4 & 1 & 0 & 0\\ 2 & -3 & 4 & 0 & 1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{array}\right)$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 2 & -3 & 4 & 0 & 1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{array}\right)\left[R_1'=\frac{R_1}{3}\right]$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 0 & -1 & \frac{4}{3} & -\frac{2}{3} & 1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{array}\right)\left[R_2'=R_2-2R_1\right]$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & -\frac{4}{3} & \frac{1}{3} & 0 & 0\\ 0 & 1 & -\frac{4}{3} & \frac{2}{3} & -1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{array}\right)\left[R_2'=-1\times R_2\right]$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 0 & 1 & -\frac{4}{3} & \frac{2}{3} & -1 & 0\\ 0 & 0 & \frac{7}{3} & -\frac{2}{3} & 1 & \end{array}\right)\left[R_3'=R_3-R_2\right]$$

$$\Rightarrow\left(\begin{array}{ccc|ccc}1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0\\ 0 & 1 & -\frac{4}{3} & \frac{2}{7} & -1 & 0\\ 0 & 0 & 1 & -\frac{2}{7} & \frac{3}{7} & \frac{3}{7}\end{array}\right)\left[R_3'=R_3\times\frac{3}{7}\right]$$

$$\Rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} & -1 & 0 \\ 0 & 0 & 1 & -\frac{2}{7} & \frac{3}{7} & \frac{3}{7} \end{array}\right) \quad [R_1' = R_1 + R_2]$$

$$\Rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & \frac{2}{7} & -\frac{3}{7} & \frac{4}{7} \\ 0 & 0 & 1 & -\frac{2}{7} & \frac{3}{7} & \frac{3}{7} \end{array}\right) \quad [R_2' = R_2 + R_3 \times \frac{4}{3}]$$

$\rightarrow Ax = b$

$\Rightarrow A^{-1}(Ax) = A^{-1}b$

$\Rightarrow x = A^{-1}b$

$$\therefore A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ \frac{2}{7} & -\frac{3}{7} & \frac{4}{7} \\ -\frac{2}{7} & \frac{3}{7} & \frac{3}{7} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ \frac{2}{7} & -\frac{3}{7} & \frac{4}{7} \\ -\frac{2}{7} & \frac{3}{7} & \frac{3}{7} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$



$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 - 2 + 0 \\ -\frac{2}{7} - \frac{6}{7} + \frac{4}{7} \\ -\frac{2}{7} + \frac{6}{7} + \frac{3}{7} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

H.W

(1) $5x - 7y + z = 11$
$6x - 8y - z = 15$
$3x + 2y - 6z = 7$

(2) $5x - 6y + 4z = 15$
$7x + 4y - 3z = 19$
$2x + y + 6z = 16$

H·W (starts here)

① $5x - 7y + z = 11$
$6x - 8y - z = 15$
$3x + 2y - 6z = 7$

Here,
$A = \begin{bmatrix} 5 & -7 & 1 \\ 6 & -8 & -1 \\ 3 & 2 & -6 \end{bmatrix}$, $\underline{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $\underline{b} = \begin{bmatrix} 11 \\ 15 \\ 7 \end{bmatrix}$

$\therefore [A \mid I]$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 5 & -7 & 1 & 1 & 0 & 0 \\ 6 & -8 & -1 & 0 & 1 & 0 \\ 3 & 2 & -6 & 0 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 1 & 0 \\ 6 & -8 & -1 & 0 & 1 & 0 \\ 3 & 2 & -6 & 0 & 0 & 1 \end{array}\right] \quad [R_1' = R_2 - R_1]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 1 & 0 \\ 0 & -2 & 11 & 6 & -5 & 0 \\ 0 & 5 & 0 & 3 & -3 & 1 \end{array}\right] \quad \begin{bmatrix} R_2' = R_2 - 6R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 1 & 0 \\ 0 & 5 & 0 & 3 & -3 & 1 \\ 0 & -2 & 11 & 6 & -5 & 0 \end{array}\right] \quad [R_2 \leftrightarrow R_3]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & \frac{3}{5} & \frac{-3}{5} & \frac{1}{5} \\ 0 & -2 & 11 & 6 & -5 & 0 \end{array}\right] \quad [R_2' = \frac{R_2}{5}]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -\frac{2}{5} & \frac{2}{5} & \frac{1}{5} \\ 0 & 1 & 0 & \frac{3}{5} & -\frac{3}{5} & \frac{1}{5} \\ 0 & 0 & 11 & 36/5 & -\frac{3}{5} & \frac{2}{5} \end{array}\right] \quad \begin{bmatrix} R_1' = R_1 + R_2 \\ R_3' = R_3 + 2R_2 \end{bmatrix}$$

H.W

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -2/5 & 2/5 & 1/5 \\ 0 & 1 & 0 & 3/5 & -3/5 & 1/5 \\ 0 & 0 & 1 & 36/55 & -31/55 & 2/55 \end{array}\right] \quad \left[R_3' = \frac{R_3}{11}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 10/11 & -8/11 & 3/11 \\ 0 & 1 & 0 & 3/5 & -3/5 & 1/5 \\ 0 & 0 & 1 & 36/55 & -31/55 & 2/55 \end{array}\right] \quad \left[R_1' = R_1 + 2R_3\right]$$

$$A^{-1} = \begin{bmatrix} 10/11 & -8/11 & 3/11 \\ 3/5 & -3/5 & 1/5 \\ 36/55 & -31/55 & 2/55 \end{bmatrix}$$

$\rightarrow A\underline{x} = \underline{b}$

$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}\underline{b}$

$\Rightarrow \underline{x} = A^{-1}\underline{b}$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10/11 & -8/11 & 3/11 \\ 3/5 & -3/5 & 1/5 \\ 36/55 & -31/55 & 2/55 \end{bmatrix} \begin{bmatrix} 11 \\ 15 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

H.W

② $5x-6y+4z=15$
$7x+4y-3z=19$
$2x+y+6z=46$

Here, $A=\begin{bmatrix}5 & -6 & 4\\ 7 & 4 & -3\\ 2 & 1 & 6\end{bmatrix}$, $x=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$, $b=\begin{bmatrix}15\\ 19\\ 46\end{bmatrix}$

$\therefore [A|I]$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 5 & -6 & 4 & 1 & 0 & 0\\ 7 & 4 & -3 & 0 & 1 & 0\\ 2 & 1 & 6 & 0 & 0 & 1\end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -8 & -8 & 1 & 0 & -2\\ 7 & 4 & -3 & 0 & 1 & 0\\ 2 & 1 & 6 & 0 & 0 & 1\end{array}\right] \quad [R_1' = R_1 - 2R_3]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -8 & -8 & 1 & 0 & -2\\ 0 & 60 & 53 & -7 & 1 & 14\\ 0 & 17 & 22 & -2 & 0 & 5\end{array}\right] \quad \begin{bmatrix}R_2' = R_2 - 7R_1\\ R_3' = R_3 - 2R_1\end{bmatrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -8 & -8 & 1 & 0 & -2\\ 0 & 1 & 53/60 & -7/60 & 1/60 & 14/60\\ 0 & 17 & 22 & -2 & 0 & 5\end{array}\right] \quad \left[R_2' = \frac{-R_2}{60}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -14/15 & 1/15 & 2/15 & -2/15\\ 0 & 1 & 53/60 & -7/60 & 1/60 & 14/60\\ 0 & 0 & 419/60 & -1/60 & -17/60 & 31/30\end{array}\right] \quad \begin{bmatrix}R_1' = R_1 + 8R_2\\ R_3' = R_3 - 17R_2\end{bmatrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -14/15 & 1/15 & 2/15 & -2/15\\ 0 & 1 & 53/60 & -7/60 & 1/60 & 14/60\\ 0 & 0 & 1 & -1/419 & -17/419 & 62/419\end{array}\right] \quad \left[R_3' = \frac{60}{419}R_3\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 27/419 & 40/419 & 2/419\\ 0 & 1 & 0 & -48/419 & 22/419 & 43/419\\ 0 & 0 & 1 & -1/419 & -17/419 & 62/419\end{array}\right] \quad \begin{bmatrix}R_1' = R_1 + \frac{14}{15}R_3\\ R_2' = R_2 - \frac{53}{60}R_3\end{bmatrix}$$

$\sim [I|A^{-1}]$

H.w (Ends here)

$$\therefore A^{-1} = \begin{bmatrix} \frac{27}{419} & \frac{40}{419} & \frac{2}{419} \\ -\frac{48}{419} & \frac{22}{419} & \frac{43}{419} \\ -\frac{1}{419} & \frac{-17}{419} & \frac{62}{419} \end{bmatrix}$$

$\rightarrow A\underline{x} = \underline{b}$

$\Rightarrow A^{-1}(A\underline{x}) = A^{-1}\underline{b}$

$\Rightarrow \underline{x} = A^{-1}\underline{b}$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 27/419 & 40/419 & 2/419 \\ -48/419 & 22/419 & 43/419 \\ -1/419 & -17/419 & 62/419 \end{bmatrix} \begin{bmatrix} 15 \\ 19 \\ 96 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{405}{419} + \frac{760}{419} + \frac{92}{419} \\ -\frac{720}{419} + \frac{418}{419} + \frac{1978}{419} \\ -\frac{15}{419} - \frac{323}{419} + \frac{2852}{419} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

27/9/2026

Solution of Simultaneous Linear equation by determinants [Cramner's Rule]

Let us solve the following equations:

$$a_1x+b_1y+c_1z=d_1$$
$$a_2x+b_2y+c_2z=d_2$$
$$a_3x+b_3y+c_3z=d_3$$

$$D=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$$

← 0 হলে Cramner's Rule দিয়ে করা যাবেনা

$$D_1=\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

$$D_2=\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$D_3=\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ d_3 & b_3 & d_3 \end{vmatrix}$$

$$x=\frac{D_1}{D}$$

$$y=\frac{D_2}{D}$$

$$z=\frac{D_3}{D}$$

$5x - 7y + z = 11$

$6x - 8y - z = 15$

$3x + 2y - 6z = 7$

Soln:-

From the given equations

$$D = \begin{vmatrix} 5 & -7 & 1 \\ 6 & -8 & -1 \\ 3 & 2 & -6 \end{vmatrix} = 55$$

$$D_1 = \begin{vmatrix} 11 & -7 & 1 \\ 15 & -8 & -1 \\ 7 & 2 & -6 \end{vmatrix} = 55$$

$$D_2 = \begin{vmatrix} 5 & 11 & 1 \\ 6 & 15 & -1 \\ 3 & 7 & 6 \end{vmatrix} = -55$$

$$D_3 = \begin{vmatrix} 5 & -7 & 11 \\ 6 & -8 & 15 \\ 3 & 2 & 7 \end{vmatrix} = -55$$

By cramer's Rule,

$$x = \frac{D_1}{D} = \frac{55}{55} = 1$$

$$y = \frac{D_2}{D} = \frac{-55}{55} = -1$$

$$z = \frac{D_3}{D} = \frac{-55}{55} = -1$$

Hence, $x = 1$, $y = -1$, $z = -1$

Ex:- Solve the following system of equations by using determinants

$$x+y+z=1$$
$$ax+by+cz=k$$
$$a^2x+b^2y+c^2z=k^2$$

Soln:

We have $D=\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$$=\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix} \quad \left[\begin{matrix} C_2' = C_2 - C_1 \\ C_3' = C_3 - C_1 \end{matrix}\right]$$

$$=\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & (b+a)(b-a) & (c+a)(c-a) \end{vmatrix}$$

$$=(b-a)(c-a)\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 1 \\ a^2 & b+a & c+a \end{vmatrix}$$

$$=(b-a)(c-a)\cdot 1\begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix}$$

$$=(b-a)(c-a)(c+a-b-a)$$
$$=(b-a)(c-a)(c-b)$$
$$=(b-c)(c-a)(a-b)$$

Now,

$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ k & b & c \\ k^2 & b^2 & c^2 \end{vmatrix}$$ ~~[Replacing~~

~~Replacing a by~~

Replacing a by k in (1)

$$D_1 = (b-c)(c-k)(k-b)$$

$$D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & k & c \\ a^2 & k^2 & c^2 \end{vmatrix}$$

Replacing b by k in (1), $D_2 = (k-c)(c-a)(a-k)$

$$D_3 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & k \\ a^2 & b^2 & k^2 \end{vmatrix}$$

Replacing c by k in (1), $D_3 = (a-b)(b-k)(k-a)$

$$\therefore x = \frac{D_1}{D} = \frac{(b-c)(c-k)(k-b)}{(b-c)(c-a)(a-b)} = \frac{(c-k)(k-b)}{(c-a)(a-b)}$$

$$\therefore y = \frac{D_2}{D} = \frac{(k-c)(c-a)(a-k)}{(b-c)(c-a)(a-b)} = \frac{(k-c)(a-k)}{(b-c)(a-b)}$$

$$\therefore z = \frac{D_3}{D} = \frac{(a-b)(b-k)(k-a)}{(b-c)(c-a)(a-b)} = \frac{(b-k)(k-a)}{(b-c)(c-a)}$$

Determinant থেকে 10 marks এর একটি Determinant থেকে final এ এসেছে।

Ex:-
The sum of three numbers is 6. If we multiply the third number by 2 and and add the first number to the result, we get 7. By adding second and third numbers to three times the first number we get 12. Using determinant find the numbers.

Soln:-
Let the three numbers be x, y and z
Then from the given conditions, we have,

$$x+y+z=6$$
$$x+2z=7$$
$$3x+y+z=12$$

Here,

$$D=\begin{vmatrix}1 & 1 & 1\\ 1 & 0 & 2\\ 3 & 1 & 1\end{vmatrix}=4$$

$$D_1=\begin{vmatrix}6 & 1 & 1\\ 7 & 0 & 2\\ 12 & 1 & 1\end{vmatrix}=12$$

$$D_2=\begin{vmatrix}1 & 6 & 1\\ 1 & 7 & 2\\ 3 & 12 & 1\end{vmatrix}=4$$

$$D_3=\begin{vmatrix}1 & 1 & 6\\ 1 & 0 & 7\\ 3 & 1 & 12\end{vmatrix}=8$$

$$\therefore x=\frac{D_1}{D}=\frac{12}{4}=3$$

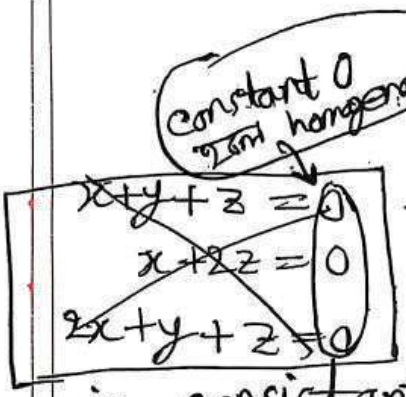
$$\therefore y=\frac{D_2}{D}=\frac{4}{4}=1 \qquad z=\frac{D_3}{D}=\frac{8}{4}=2$$

Thus the three numbers are 3, 1, 2

Homogeneous System

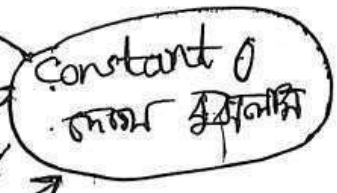
$D = 0$
Consistent with infinitely many solutions.

$D \neq 0$
Consistant with unique solution
$x=0, y=0, z=0$
(Trivial Solution)



Ex:
Find the values of λ, for which the following system of equations is consistant and has non-trivial solution.

$$(\lambda-1)x + (3\lambda+1)y + 2\lambda z = 0$$
$$(\lambda-1)x + (4\lambda-2)y + (\lambda+3)z = 0$$
$$2x + (3\lambda+1) + 3(\lambda-1)z = 0$$



Solution:
This is a system of homogeneous equations.
For non-trivial solution, $D = 0$.

$$\therefore D = \begin{vmatrix} \lambda-1 & 3\lambda+1 & 2\lambda \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 0 & -\lambda+3 & \lambda-3 \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0 \quad [R_1' = R_1 - R_2]$$

$$\Rightarrow \begin{vmatrix} 0 & 0 & \lambda-3 \\ \lambda-1 & 5\lambda+1 & \lambda+3 \\ 2 & 6\lambda-2 & 3\lambda-3 \end{vmatrix} = 0 \quad [C_2' = C_2 + C_3]$$

$$\Rightarrow (\lambda-3)\left[(\lambda-1)(6\lambda-2) - 2(5\lambda+1)\right] = 0$$

$$\Rightarrow \left[6\lambda^2 - 8\lambda + 2 - 10\lambda - 2\right] = 0$$

$$\Rightarrow 6\lambda^2 - 18\lambda = 0$$

$$\Rightarrow 6\lambda(\lambda-3) = 0$$

$$\therefore \lambda = 3, 0$$

29/9/2026

A system of linear equation has either

① no solution
② Exactly one solution
③ infinitely many solutions

⊛ A system of linear equations is called inconsistent if it has no solution

⊛ A system of linear equations is said to be consistent if it has either one solution or infinitely many solution

Gaussian Elimination method

Consider

$$a_1x_1 + b_1x_2 + c_1x_3 = d_1$$
$$a_2x_1 + b_2x_2 + c_2x_3 = d_2$$
$$a_3x_1 + b_3x_2 + c_3x_3 = d_3$$

⊛ Step 1: Form the argmented matrix

$$\left(\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array}\right)$$

Step 2: Form the row echelon form by elementary row operation

$$\left(\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ 0 & b_2' & c_2' & d_2' \\ 0 & 0 & c_3' & d_3' \end{array}\right)$$

Step 3:-

In equation form,

$$a_1 x_1 + b_1 x_2 + c_1 x_3 = d_1$$
$$b_2' x_2 + c_2' x_3 = d_2'$$
$$c_3' x_3 = d_3'$$

Which is the $A\underline{x} = \underline{b}$ form

where, $A = \begin{pmatrix} a_1 & b_1 & c_1 \\ 0 & b_2' & c_2' \\ 0 & 0 & c_3' \end{pmatrix}$, $\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$; $\underline{b} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$

Step 4:-

By back substitution we get the result

① $x+y+z=1$

$3x-y-z=4$

$x+5y+5z=-1$

Soln:-

The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 3 & -1 & -1 & 4 \\ 1 & 5 & 5 & -1 \end{array}\right)$$

By elementary row operation,

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & -4 & 1 \\ 0 & 4 & 4 & -2 \end{array}\right) \begin{bmatrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & -4 & 1 \\ 0 & 0 & 0 & -1 \end{array}\right) \quad [R_3' = R_3 + R_2]$$

In matrix form,

$$x+y+z=1$$
$$-4y-4z=1$$
$$0=-1$$

$0=-1$ is not true

The system is inconsistent

Thus there is no solution for this system.

অগ্রাধিকার না, যেমন এভাবেও করা যায়

② $2x+y-2z=10$
$3x+2y+2z=1$
$5x+4y+3z=4$

Soln:-
Reduce the system to echelon form by means of elementary operation,

$$\sim \begin{cases} 2x+y-2z=10 \\ y+10z=-28 \\ 3y+16z=-42 \end{cases} \quad \begin{bmatrix} L_2' = 2L_2 - 3L_1 \\ L_3' = 2L_3 - 5L_1 \end{bmatrix}$$

↖ L means line

$$\sim \begin{cases} 2x+y-2z=10 \\ y+10z=-28 \\ -14z=42 \end{cases} \quad [L_3' = L_3 - 3L_2]$$

By back substitution we get

$-14z = 42$
$\therefore z = -3$

$y = -10z - 28 = -10(-3) - 28 = 2$

$2x = -y + 2z + 10 = -2 - 6 + 10 = -12 + 10 = -2$
$\therefore x = -1$

Ans: $(-1, 2, -3)$

System solution এর উদাহরণ CT3 2021।

~~System সমাধানের ধাপ~~

③ $x+2y-3z=-2$
$3x-y-2z=1$
$2x+3y-5z=-3$

Soln:-

The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 3 & -1 & -2 & 1 \\ 2 & 3 & -5 & -3 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & -7 & 7 & 7 \\ 0 & -1 & 1 & 1 \end{array}\right) \begin{bmatrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 2R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & 1 & -1 & -1 \\ 0 & -1 & 1 & 1 \end{array}\right) \left[R_2' = -\frac{1}{7} R_2\right]$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right) \left[R_3' = R_2 + R_3\right]$$

The corresponding system is

$$x+2y-2z=-2$$
$$y-z=-1$$

Thus system has two equations and three variable. The free variable $= (3-2)=1$.
Let the free variable be z
Let $z=5$
$\therefore y=4, x=5$

~~particular solution / Infinitely many solution~~
particular solution $(5,4,5)$

parametric solution

Let $z = t$,

$\therefore y = -1 + t$

$\therefore x = -2 - 2y + 3z$

$= -2 - 2(-1+t) + 3t$

$= -2 + t - 2t + 3t$

$= t$

$\therefore (t, (-1+t), t)$

C.W

① $2x - 3y + 5z = 1$

$3x + y - z = 2$

$x + 4y - 6z = 1$

② $x - 4y + 5z = 8$

$3x + 7y - z = 3$

$x + 15y - 11z = -14$

③ $x + 2y + 3z = 0$

$2x + 5y + 2z = 0$

$3x - y - 4z = 0$

H.W.

~~Exercises~~

① $2x-3y+5z=1$

$3x+y-z=2$

$x+4y-6z=1$

We have, $D=\begin{vmatrix} 2 & -3 & 5 \\ 3 & 1 & -1 \\ 1 & 4 & -6 \end{vmatrix}=0$ (সমাধান হবে না)

Reduce the system to echelon form by means of elementary operation

$$\sim \begin{cases} 2x-3y+5z=1 \\ 11y-17z=1 \\ 11y-17z=1 \end{cases} \quad \begin{bmatrix} L_2'=2L_2-3L_1 \\ L_3'=2L_3-L_1 \end{bmatrix}$$

$$\sim \begin{cases} 2x-3y+5z=1 \\ 11y-17z=1 \\ 0=0 \end{cases} \quad [L_3'=L_3-L_2]$$

The corresponding system is

$$2x-3y+5z=1$$
$$11y-17z=1$$

Thus the system has two equations and three variable

The free variable $=(3-2)=1$

Let the free variable be z

Let $z=4$

$\therefore y=6.2727272727\ldots$, $z=-0.0909090909\ldots$

particular solution $(-0.09, 6.27, 4)$

Parametric solution

Let $z=t$

$\therefore y=\frac{1+17t}{11}$

$2x=3y-5z+1=3\left(\frac{1+17t}{11}\right)-5z+1$

$\Rightarrow x=\frac{3+51t-55z+11}{22}$

$=\frac{14-4t}{22}=\frac{7-2t}{11}$

$\therefore \left(\frac{14-4t}{22}, \frac{1+17t}{11}, t\right)$

H.W

(2) $x-4y+5z=8$

$3x+7y-z=3$

$x+15y-11z=-19$

Reduce the system to echelon form by means of elementary operation

$$\sim \begin{cases} x-4y+5z=8 \\ 19y-16z=-21 \\ 19y-16z=-22 \end{cases} \quad \begin{bmatrix} L_2' = L_2 - 9L_1 \\ L_3' = L_3 - L_1 \end{bmatrix}$$

$$\sim \begin{cases} x-4y+5z=8 \\ 19y-16z=-21 \\ 0=-1 \end{cases} \quad \begin{bmatrix} L_3' = L_3 - L_2 \end{bmatrix}$$

$0=-1$ is not true

the system is inconsistent

Thus there is no solution for this system

H.W

③ $x+2y+3z=0$
$2x+5y+2z=0$
$3x-y-4z=0$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} x+2y+3z=0 \\ y-4z=0 \\ -7y-13z=0 \end{cases} \quad \begin{bmatrix} R_2'=R_2-2R_1 \\ R_3'=R_3-3R_1 \end{bmatrix}$$

$$\sim \begin{cases} x+2y+3z=0 \\ y-4z=0 \\ -41z=0 \end{cases} \quad \left[R_3'=R_3+7R_2\right]$$

By back substitution we get

$-41z=0$

$\Rightarrow z=0$

$y-4\times 0=0$

$\Rightarrow y=0$

$x+2\times 0+3\times 0=0$

$\Rightarrow x=0$

Ans $(0,0,0)$

A/ 10/26

Solve the following system

(1) $2x+4y+6z=22$
$3x+8y+5z=27$
$-x+y+2z=2$

Ans: $(3, 1, 2)$

(2) $3x-2y+8z=9$
$-2x+2y+z=3$
$x+2y-3z=8$

Ans $(3, 4, 1)$

(3) Find the parametric solution of

$x+2y-3z=-2$
$3x-y-2z=1$
$2x+3y-5z=-3$

Ans: $\{(t, t-1, t): t \in \mathbb{R}$

(4) $x-4y+5z=8$
$3x+7y-z=3$
$x+15y-11z=-14$

No solution

(5) Find the particular solution of

$x_1+2x_2-3x_3=6$
$2x_1-x_2+4x_3=2$
$4x_1+3x_2-2x_3=14$

① $2x+4y+6z=22$
$3x+8y+5z=27$
$-x+y+2z=2$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} 2x+4y+6z=22 \\ 4y-8z=-12 \\ 6y+10z=26 \end{cases} \quad [R_2' = 2R_2 - 3R_1,\ R_3' = 2R_3 + R_1]$$

$$\sim \begin{cases} 2x+4y+6z=22 \\ 4y-8z=-12 \\ 44z=88 \end{cases} \quad [R_3' = 2R_3 - 3R_2]$$

By back substitution we get

$44z=88$

$\Rightarrow z=\frac{88}{44}=2$

$4y-8\times 2=-12$

$\Rightarrow 4y=-12+16$

$\Rightarrow y=1$

$2x+4\times 1+6\times 2=22$

$\Rightarrow 2x+16=22$

$\Rightarrow 2x=6$

$\Rightarrow x=3$

$\therefore$ Ans $=(3,1,2)$

② $3x-2y+8z=9$

$-2x+2y+z=3$

$x+2y-3z=8$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} 3x-2y+8z=9 \\ 2y+19z=27 \\ 8y-17z=15 \end{cases} \quad \begin{bmatrix} R_2' = 3R_2+2R_1 \\ R_3' = 3R_3-R_1 \end{bmatrix}$$

$$\sim \begin{cases} 3x-2y+8z=9 \\ 2y+19z=27 \\ -93z=-93 \end{cases} \quad [R_3' = R_3 - 4R_2]$$

By back substitution we get,

$-93z=-93$

$\Rightarrow z=\frac{-93}{-93}=1$

$2y+19\times1=27$

$\Rightarrow 2y=27-19=8$

$\Rightarrow y=4$

$3x-2\times4+8\times1=9$

$\Rightarrow 3x=9+8-8$

$\Rightarrow x=3$

$\therefore$ Ans $=(3,4,1)$

③ $x+2y-3z=-2$

$3x-y-2z=1$

$2x+3y-5z=-3$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} x+2y-3z=-2 \\ -7y+7z=7 \\ -y+z=1 \end{cases} \quad \left[\begin{array}{l} L_2' = L_2 - 3L_1 \\ L_3' = L_3 - 2L_1 \end{array}\right]$$

$$\sim \begin{cases} x+2y-3z=-2 \\ -7y+7z=7 \\ 0=0 \end{cases} \quad \left[L_3' = 7L_3 - L_2\right]$$

The corresponding system is

$$x+2y-3z=-2$$
$$-7y+7z=7$$

Thus the system has two equation and three variable.

The free variable $=(3-2)=1$

Let $z=t$

$\therefore -7y+7t=7$

$\Rightarrow -y+t=1$

$\Rightarrow y=t-1$

$\therefore x+2(t-1)-3t=-2$

$\Rightarrow x+2t-2-3t=-2$

$\Rightarrow x-t=0$

$\Rightarrow x=t$

$\therefore$ Ans $(t, t-1, t)$

④ $x-4y+5z=8$

$3x+7y-z=3$

$x+15y-11z=-14$

Reduce the system to echelon form by means of elementary operations

$$\sim \begin{cases} x-4y+5z=8 \\ 19y-16z=-21 \\ 19y-16z=-22 \end{cases} \quad \left[\begin{matrix} R_2'=R_2-3R_1 \\ R_3'=R_3-R_1 \end{matrix}\right] \quad \left[\begin{matrix} L_2'=L_2-3L_1 \\ L_3'=L_3-L_1 \end{matrix}\right]$$

$$\sim \begin{cases} x-4y+5z=8 \\ 19y-16z=-21 \\ 0=-1 \end{cases} \quad \left[\begin{matrix} R_3'=R_3-R_2 \\ L_3'=L_3-L_2 \end{matrix}\right]$$

$0=-1$ is not true

∴ The system is inconsistent

Thus there is no solution for this

⑤ $x_1 + 2x_2 - 3x_3 = 6$

$2x_1 - x_2 + 4x_3 = 2$

$4x_1 + 3x_2 - 2x_3 = 14$

$$\sim \begin{cases} x_1 + 2x_2 - 3x_3 = 6 \\ -5x_2 + 10x_3 = -10 \\ -5x_2 + 10x_3 = -10 \end{cases} \left[\begin{array}{l} L_2' = L_2 - 2L_1 \\ L_3' = L_3 - 4L_1 \end{array} \right]$$

$$\sim \begin{cases} x_1 + 2x_2 - 3x_3 = 6 \\ -5x_2 + 10x_3 = -10 \\ 0 = 0 \end{cases} \left[L_3' = L_3 - L_2 \right]$$

The corresponding system is

$$x_1 + 2x_2 - 3x_3 = 6$$
$$-5x_2 + 10x_3 = -10$$

Thus system has two equations and three variable. The free variable $= (3-2) = 1$

Let $x_3 = 3$

$\therefore -5x_2 + 10 \times 3 = -10$

$\Rightarrow -5x_2 = -10 - 30$

$\therefore x_2 = 8$

$x_1 + 2 \times 8 - 3 \times 3 = 6$

$x_1 + 16 - 9 = 6$

$\Rightarrow x_1 = -1$

$\therefore$ Particular solution $(-1, 8, 3)$

6/10/2026 কাল final ও আজকে CT তে না

We call $\underline{b} - A\underline{x}$ the least squares error vector
we call $\|\underline{b} - A\underline{x}\|$ the least squares error

(*) Find the least squares solution/ the least squares error vector/ the least squares error of the linear system.

$$x_1 - x_2 = 4$$
$$3x_1 + 2x_2 = 1$$
$$-2x_1 + 4x_2 = 3$$

Solution:

It will be convenient to express the system in the matrix form $A\underline{x} = \underline{b}$, where

$$A = \begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix} \text{ and } \underline{b} = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}$$

It follows that

$$A^TA = \begin{pmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 14 & 3 \\ -3 & 21 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{pmatrix}\begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \end{pmatrix}$$

So the normal system $A^TA\underline{x} = A^T\underline{b}$ is

$$\begin{pmatrix} 14 & -3 \\ -3 & 21 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \end{pmatrix}$$

Solving this system yields an unique least squares solution, namely,

$$x_1 = \frac{17}{95}, \quad x_2 = \frac{143}{285}$$

The least squares error vector is

$$\underline{b} - A\underline{x}$$

$$= \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} \frac{14}{95} \\ \frac{143}{285} \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} -\frac{92}{285} \\ \frac{439}{285} \\ \frac{95}{57} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1232}{285} \\ -\frac{154}{285} \\ \frac{4}{3} \end{pmatrix}$$

and the least squares error is

$$\|b - A\underline{x}\| = \sqrt{\left(\frac{1232}{285}\right)^2 + \left(-\frac{154}{285}\right)^2 + \left(\frac{4}{3}\right)^2}$$

$$= 4.556$$

(*) Find the least square solution of $A\underline{x} = \underline{b}$

(1) $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$

(2) $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \\ 3 & 1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

(3) $A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix}$

(4) $A = \begin{pmatrix} 2 & 0 & -1 \\ 1 & -2 & 2 \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix}$

(1) $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$

It follows that

$$A^TA = \begin{pmatrix} 1 & 2 & 4 \\ -1 & 3 & 5 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 21 & 25 \\ 25 & 35 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 1 & 2 & 4 \\ -1 & 3 & 5 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} = \begin{pmatrix} 20 \\ 20 \end{pmatrix}$$

So the normal system ~~yields an unique least~~

$A^TA\underline{x} = A^T\underline{b}$ is

$$\begin{pmatrix} 21 & 25 \\ 25 & 35 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 20 \\ 20 \end{pmatrix}$$

solving the system yields an unique least squares solution namely,

$$x_1 = \frac{20}{11}, \quad x_2 = \frac{-8}{11}$$

The least squares error vector is

$$\underline{b} - A\underline{x}$$

$$= \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix}\begin{pmatrix} \frac{20}{11} \\ -\frac{8}{11} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{6}{11} \\ -\frac{27}{11} \\ \frac{15}{11} \end{pmatrix}$$

and the least squares error is

$$\|\underline{b} - A\underline{x}\| = \sqrt{\left(\frac{-6}{11}\right)^2 + \left(-\frac{27}{11}\right)^2 + \left(\frac{15}{11}\right)^2}$$

$$= \frac{3\sqrt{110}}{11} = 2.86$$

② $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \\ 3 & 1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

It follows that

$$A^TA = \begin{pmatrix} 2 & 1 & 3 \\ -2 & 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & -2 \\ 1 & 1 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 14 & 0 \\ 0 & 6 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 2 & 1 & 3 \\ -2 & 1 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$$

So the normal system $A^TA\underline{x} = A^T\underline{b}$ is

$$\begin{pmatrix} 14 & 0 \\ 0 & 6 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix},$$

Solving the system yield an unique least squares solution namely

$$\therefore x_1 = \frac{3}{7}, \; x_2 = -\frac{2}{3}$$

The least squares error vector is

$$\underline{b} - A\underline{x} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 & -2 \\ 1 & 1 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} \frac{3}{7} \\ -\frac{2}{3} \end{pmatrix} = \begin{pmatrix} -\frac{4}{21} \\ -\frac{16}{21} \\ \frac{8}{21} \end{pmatrix}$$

And the least squares error is

$$\|\underline{b} - A\underline{x}\| = \sqrt{\left(-\frac{4}{21}\right)^2 + \left(\frac{-16}{21}\right)^2 + \left(\frac{8}{21}\right)^2}$$

$$= \sqrt{\frac{16+256+64}{441}}$$

$$= \sqrt{\frac{336}{441}}$$

$$= \frac{4\sqrt{21}}{21} = 0.87$$

Work at home

③ $A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix}$

It follows that

$$A^TA = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & -2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+4+1+1 & 0+2+1+1 & -1-4+0-1 \\ 0+2+1+1 & 0+1+1+1 & 0-2+0-1 \\ -1-4+0-1 & 0-2+0-1 & 1+4+0+1 \end{pmatrix}$$

$$= \begin{pmatrix} 7 & 4 & -6 \\ 4 & 3 & -3 \\ -6 & -3 & 6 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & -2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix} = \begin{pmatrix} 6+0+9+3 \\ 0+0+9+3 \\ -6-0+0-3 \end{pmatrix} = \begin{pmatrix} 18 \\ 12 \\ -9 \end{pmatrix}$$

So the normal system $A^TA\underline{x} = A^Tb$ is

$$\begin{pmatrix} 7 & 4 & -6 \\ 4 & 3 & -3 \\ -6 & -3 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 18 \\ 12 \\ -9 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 7x_1 + 4x_2 - 6x_3 = 18 \\ 4x_1 + 3x_2 - 3x_3 = 12 \\ -6x_1 - 3x_2 + 6x_3 = -9 \end{cases}$$

Solving the system yield an unique least squares solution namely

$$x_1 = 12,\ x_2 = -3,\ x_3 = 9$$

The least squares error vector is

$$\underline{b} - A\underline{x} = \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 12 \\ -3 \\ 9 \end{pmatrix}.$$

$$= \begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix} - \begin{pmatrix} 12+0-9 \\ 24-3-18 \\ 12-3+0 \\ 12-3-9 \end{pmatrix}$$

$$\begin{pmatrix} 6 \\ 0 \\ 9 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3 \\ 9 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ -3 \\ 0 \\ 3 \end{pmatrix}$$

and the least squares error is

$$\|\underline{b} - A\underline{x}\| = \sqrt{3^2+(-3)^2+0^2+3^2}$$

$$= \sqrt{9+9+0+9}$$

$$= \sqrt{27}$$

$$= 5.2$$

(4) Given,

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 1 & -2 & 2 \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad \underline{b} = \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix}$$

It follows that

$$A^TA = \begin{pmatrix} 2 & 1 & 2 & 0 \\ 0 & -2 & -1 & 1 \\ -1 & 2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 & -1 \\ 1 & -2 & 2 \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 4+1+4+0 & 0-2-2+0 & -2+2+0+0 \\ 0-2-2+0 & 0+4+1+1 & +0-1-0-1 \\ -2+2+0-0 & -0-1-0-1 & 1+1+0+1 \end{pmatrix}$$

$$= \begin{pmatrix} 9 & -4 & 0 \\ -4 & 6 & -5 \\ 0 & -5 & 6 \end{pmatrix}$$

$$A^T\underline{b} = \begin{pmatrix} 2 & 1 & 2 & 0 \\ 0 & -2 & -1 & 1 \\ -1 & 2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0+6+0+0 \\ 0-12+0+0 \\ 0+12+0+0 \end{pmatrix} = \begin{pmatrix} 6 \\ -12 \\ 12 \end{pmatrix}$$

So the normal system $A^TA\underline{x} = A^T\underline{b}$ is

$$\begin{pmatrix} 9 & -4 & 0 \\ -4 & 6 & -5 \\ 0 & -5 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 6 \\ -12 \\ 12 \end{pmatrix}$$

$$= \begin{cases} 9x_1 - 4x_2 = 6 \\ -4x_1 + 6x_2 - 5x_3 = -12 \\ -5x_2 + 6x_3 = 12 \end{cases}$$

Solving the system yield an unique least squares solution namely

$$x_1 = 6, \; x_2 = 12, \; x_3 = 12$$

The least error vector is

$$\underline{b} - A\underline{x} = \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 & 0 & -1 \\ 1 & -2 & 2 \\ 3 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 6 \\ 12 \\ 12 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 12+0-12 \\ 6-24+24 \\ 12-12+0 \\ 0+12-12 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

and the least squares error is

$$\|b - A\underline{x}\| = \sqrt{0^2+0^2+0^2+0^2} = 0$$

$5x - 7y + z = 11$

$6x - 8y - z = 15$

$3x + 2y - 6z = 7$

Reduce the system to echelon form by means of elementary operations -

$$\sim \begin{cases} 5x - 7y + z = 11 \\ 2y - 11z = 9 \\ 31y - 33z = 2 \end{cases} \quad \begin{bmatrix} L_2' = 5L_2 - 6L_1 \\ L_3' = 5L_3 - 3L_1 \end{bmatrix}$$

$$\sim \begin{cases} 5x - 7y + z = 11 \\ 2y - 11z = 9 \end{cases} \quad [L_3' = 2L_3 - 31L_2]$$

$$275z = -275$$

By back substitution we get,

$$275z = -275$$

$$\Rightarrow z = \frac{-275}{275} = -1$$

$$2y - 11z = 9$$

$$2 \times y - 11 \times (-1) = 9$$

$$\Rightarrow y = \frac{9 - 11}{2} = \frac{-2}{2} = -1$$

$$5x - 7y + z = 11$$

$$\Rightarrow 5x - 7(-1) + (-1) = 11$$

$$\Rightarrow 5x + 7 - 1 = 11$$

$$\Rightarrow 5x = 11 - 7 + 1 = 5$$

$$\Rightarrow x = 1$$

$\therefore$ Ans $(1, -1, -1)$

13/10/2026

এই বার্তা আকারে final এ ~~[illegible]~~ থেকে ৫২ টি প্রশ্ন

Linear Combination, linear dependence & Independence

*) Vector $\underline{u}$ is a linear combination of vectors $v_1, v_2, v_n \in V$ for any scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that

$\underline{u} = \alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n = 0$

has ony the trivial sol^n $\alpha_1 = 0, \alpha_2 = 0 \ldots, \alpha_n = 0$. If there are also nontrivial sol^n, then S is colled linerly dependent.

1. At $\underline{v} = (5, 9, 5)$ and $\underline{v_1} = (2, 1, 4)$, $\underline{v_2} = (1, -1, 3)$, $\underline{v_3} = (3, 2, 5)$.
Show that $\underline{v}$ can be expressed as a linear combination of vectors $\underline{v_1}$, $\underline{v_2}$ and $\underline{v_3}$.

Sol^n:-

$\underline{v}$ can be expressed as a linear combination of the vectors $\underline{v_1}$, $\underline{v_2}$ and $\underline{v_3}$ if for any scalar α_1, α_2 and α_3 such that

$$\underline{v} = \alpha_1 \underline{v_1} + \alpha_2 \underline{v_2} + \alpha_3 \underline{v_3}$$

$$\Rightarrow (5, 9, 5) = \alpha_1 (2, 1, 4) + \alpha_2 (1, -1, 3) + \alpha_3 (3, 2, 5)$$

$$\Rightarrow (5, 9, 5) = (2\alpha_1, \alpha_1, 4\alpha_1) + (\alpha_2, -\alpha_2, 3\alpha_2) + (3\alpha_3, 2\alpha_3, 5\alpha_3)$$

$$\Rightarrow (5, 9, 5) = (2\alpha_1 + \alpha_2 + 3\alpha_3, \alpha_1 - \alpha_2 + 2\alpha_3, 4\alpha_1 + 3\alpha_2 + 5\alpha_3)$$

ডিটি প্রশ্নের সমাধান H.W

Equating the corresponding components, we get the following syste of linear equations,

$$\left.\begin{aligned} 2\alpha_1 + \alpha_2 + 3\alpha_3 &= 5 \\ \alpha_1 - \alpha_2 + 2\alpha_3 &= 9 \\ 4\alpha_1 + 3\alpha_2 + 5\alpha_3 &= 5 \end{aligned}\right\} \ldots\ldots (1)$$

$$\sim \begin{cases} 2\alpha_1 + \alpha_2 + 3\alpha_3 = 5 \\ -3\alpha_2 + \alpha_3 = 13 \\ 7\alpha_2 - 3\alpha_3 = -31 \end{cases} \quad \begin{bmatrix} L_2' = 2L_2 - L_1 \\ L_3' = L_3 - 4L_2 \end{bmatrix}$$

$$\sim \begin{cases} 2\alpha_1 + \alpha_2 + 3\alpha_3 = 5 \\ -3\alpha_2 + \alpha_3 = 13 \\ -2\alpha_3 = -12 \end{cases} \quad \left[L_3' = 3L_3 + 7L_2\right]$$

Using back substitution method from 3rd equation, we get $\alpha_3 = 1$

Substituting $\alpha_3 = 1$ in 2nd equdtion, we get

$$-3\alpha_2 + 1 = 13$$

$$\Rightarrow \alpha_2 = -4$$

From I st equation

$$2\alpha_1 - 4 + 3 = 5$$

$$\Rightarrow \alpha_1 = 3$$

Thus the soln of the system is

$$\alpha_1 = 3, \quad \alpha_2 = -4, \quad \alpha_3 = 1$$

Therefore $\underline{V} = 3\underline{v_1} - 4\underline{v_2} + \underline{v_3}$

Hence $\underline{v}$ can be expressed as a linear combination of $\underline{v_1}$, $\underline{v_2}$ and $\underline{v_3}$

যদি solution না থাকে

Hence $\underline{v}$ can not be expressed as a linear combination of $\underline{v_1}$, $\underline{v_2}$ and $\underline{v_3}$ (Final প্রশ্নের ক্ষেত্রে আসবে)

25 Oct

Eigen value and Eigen Vector

1) A scalar λ is called an eigenvalue of a matrix A if there is a non-trivial solution $\underline{x}$ of $A\underline{x} = \lambda\underline{x}$ and $\underline{x}$ is called the eigen vector

2) $(\lambda I - A)$ or $(A - \lambda I)$ is called the characteristic matrix of A

3) $|\lambda I - A|$ or $|A - \lambda I|$ is called the characteristic polynomial of A

4) $|\lambda I - A| = 0$ or $|A - \lambda I| = 0$ is called the characteristic equation of A

Ex 1:
Find the characteristic matrix of

$$A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

Soln:

$$A - \lambda I$$

$$= \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{pmatrix}$$

is the characteristic matrix.

Ex 2:
Find the characteristic ~~matrix~~ polynomial of

Soln:

$$\begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix}$$

$$= (2-\lambda)(6-5\lambda+\lambda^2-2) - 2(2-\lambda+1) + 1(2-3+\lambda)$$

$$= (2-\lambda)(\lambda^2-5\lambda+4) - 2(3-\lambda) + 1(\lambda-1)$$

$$= -\lambda^3+7\lambda^2-11\lambda+5\lambda^0$$

Cubic polynomial

Ex 3:-

Find the characteristic roots or Eigen values

Soln:-

Ex 2- এর solution সম্পর্কে বিস্তারিত দেখ ——

→ $\lambda^3 + 7\lambda^2 + 11\lambda - 5 = 0$

$\Rightarrow \lambda^3 - \lambda^2 - 6\lambda^2 + 6\lambda + 5\lambda - 5 = 0$

$\Rightarrow \lambda^2(\lambda-1) - 6\lambda(\lambda-1) + 5(\lambda-1) = 0$

$\Rightarrow (\lambda-1)(\lambda^2 - 6\lambda + 5)$

$\Rightarrow (\lambda-1)(\lambda^2 - 5\lambda - \lambda + 5) = 0$

$\Rightarrow (\lambda-1)[\lambda(\lambda-5) - 1(\lambda-5)] = 0$

$\Rightarrow (\lambda-1)(\lambda-5)(\lambda-1) = 0$

$\lambda - 1 = 0 \quad \lambda - 5 = 0 \quad , \quad \lambda - 1 = 0$

$\Rightarrow \lambda = 1 \quad \Rightarrow \lambda = 5 \quad = \lambda = 1$

$\therefore \lambda = 1, 1, 5$

↑ Repeated value

যাতেনাও বিস্তারিত হবে

CT-6 প্রশ্ন

$\begin{matrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{matrix}$ 129

C.W Find the characteristic roots or Eigen values of $\begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$

$\Rightarrow A - \lambda I$

$= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$

$= \begin{pmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{pmatrix}$

$|A - \lambda I| = \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix}$

Calculation Problem

$= (6-\lambda)\{(3-\lambda)^2 - 1\} + 2[-2(3-\lambda) + 1(2)] + 2[2 - 2(3-\lambda)]$

$= (6-\lambda)(9 - 6\lambda + \lambda^2 - 1) + 2(-6 - 2\lambda + 2) + 2(2 - 6 + 2\lambda)$

$= (6-\lambda)(8 - 6\lambda + \lambda^2) - 12 - 4\lambda + 4 + 4 - 12 + 4\lambda$

$= 48 - 36\lambda + 6\lambda^2 - 8\lambda + 6\lambda^2 - \lambda^3 - 12 - 4\lambda + 8 - 12 + 4\lambda$

$= 32 - \ldots\ldots\lambda + 12\lambda^2 - \lambda^3$

$= \lambda^3 - 12\lambda^2 + 36\lambda - 32$

Characteristic equation is $[A - \lambda I] = 0$

$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$

$\Rightarrow \lambda^3 - 2\lambda - 10\lambda^2 + 20\lambda + 16\lambda - 32 = 0$

$\Rightarrow \lambda^2(\lambda - 2) - 10\lambda(\lambda - 2) + 16(\lambda - 2) = 0$

$\Rightarrow (\lambda-2)(\lambda^2-10\lambda+16)=0$

$\Rightarrow (\lambda-2)(\lambda^2-8\lambda-2\lambda+16)=0$

$\Rightarrow (\lambda-2)\{\lambda(\lambda-8)-2(\lambda-8)\}=0$

$\Rightarrow (\lambda-2)(\lambda-8)(\lambda-2)=0$

$\therefore \lambda = 2, 2, 8$

H.W Find the charateristic roots or eigen values of $\begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}$

Sol^n:

$A-\lambda I$

$= \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$

$= \begin{pmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{pmatrix}$

$|A-\lambda I| = \begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix}$

$= (1-\lambda)\{(2-\lambda)(3-\lambda)-2\} - 1\{2-2(2-\lambda)\}$

$= (1-\lambda)\{6-2\lambda-3\lambda+\lambda^2-2\} - 1(2-4+2\lambda)$

$= (1-\lambda)(4-5\lambda+\lambda^2)+2-2\lambda$

$= 4-5\lambda+\lambda^2-4\lambda+5\lambda^2-\lambda^3+2-2\lambda$

$= -\lambda^3 + 6\lambda^2 - 11\lambda + 6$

characteristic equation is $[A - \lambda I] = 0$

$-\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$

$\Rightarrow -\lambda^3 + \lambda^2 + 5\lambda^2 - 5\lambda - 4\lambda + 4 = 0$

$\Rightarrow -\lambda^2(\lambda - 1) + 5\lambda(\lambda - 1) - 4(\lambda - 1) = 0$

$\Rightarrow (\lambda - 1)(-\lambda^2 + 5\lambda - 4) = 0$

$\Rightarrow (\lambda - 1)(-\lambda^2 + \lambda + 4\lambda - 4) = 0$

$\Rightarrow (\lambda - 1)\{-\lambda(\lambda - 1) + 4(\lambda - 1)\} = 0$

$\Rightarrow (\lambda - 1)(\lambda - 1)(-\lambda + 4) = 0$

$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

$\Rightarrow -\lambda^3 + \lambda^2 + 5\lambda^2 - 5\lambda - 6\lambda + 6 = 0$

$\Rightarrow -\lambda^2(\lambda - 1) + 5\lambda(\lambda - 1) - 6(\lambda - 1) = 0$

$\Rightarrow (\lambda - 1)(-\lambda^2 + 5\lambda - 6) = 0$

$\Rightarrow (\lambda - 1)(-\lambda^2 + 3\lambda + 2\lambda - 6) = 0$

$\Rightarrow (\lambda - 1)\{-\lambda(\lambda + 3) - 2(\lambda - 3)\} = 0$

$\Rightarrow (\lambda - 1)(-\lambda - 2)(\lambda - 3) = 0.$

$\therefore \lambda = 1, 2, 3$

27/10/2026

iterative method. pdf

Final ও 10 নম্বরে আসতে পারে Jacobi Gauss

~~The Jacobi method:~~

Ex:- Apply the jacobi method to solve

$$5x_1 - 2x_2 + 3x_3 = -1$$
$$-3x_1 + 9x_2 + x_3 = 2$$
$$2x_1 - x_2 - 7x_3 = 3$$

[An iterative process involves systematically repeating a sequence ~~of~~ or formula to achieve the desired result. Through trial and error, you may get closer to a solution.]

Continue iterations until two successive approximations are identical (rounded to 3 significant digits)

$$\begin{cases} 5x_1 = -1 + 2x_2 - 3x_3 \\ 9x_2 = 2 + 3x_1 - x_3 \\ -7x_3 = 3 - 2x_1 + x_2 \\ (7x_3 = -3 + 2x_1 - x_2) \end{cases}$$

$$\therefore x_1 = \frac{1}{5}(-1 + 2x_2 - 3x_3)$$
$$x_2 = \frac{1}{9}(2 + 3x_1 - x_3)$$
$$x_3 = \frac{1}{7}(-3 + 2x_1 - x_2) \quad \text{—} (*)$$

Consider $x^{(0)} = (0, 0, 0)$

$$x_1^{(1)} = \frac{1}{5}(-1 + 0 - 0) = -\frac{1}{5} = -0.200$$
$$x_2^{(1)} = \frac{1}{9}(2 + 0 - 0) = \frac{2}{9} = 0.222$$
$$x_3^{(1)} = \frac{1}{7}(-3 + 0 - 0) = -0.428$$

$x^{(1)} = (x_1^{(1)}, x_2^{(1)}, x_3^{(1)}) = (-0.200, 0.222, -0.428)$

Now we start the 2nd iteration,

$$x_1^{(2)} = \frac{1}{5}(-1 + 2(0.222) - 3(-0.428))$$

$$= \frac{1}{5}(-1 + 0.444 + 1.284) = \frac{1}{5}(0.728) = 0.1456$$

$$x_2^{(2)} = \frac{1}{9}(2 + 3(-0.200) + 0.428)$$

$$= \frac{1}{9}(2 - 0.6 + 0.428) = \frac{1}{9}(1.828) = 0.203$$

$$x_3^{(2)} = \frac{1}{7}(-3 + 2(-0.200) - 0.222) = \frac{1}{7}(-3 - 0.4 - 0.222)$$

$$= \frac{1}{7}(-3.622)$$

$$= -0.517$$

For 3rd iteration, we will use the values of 2nd iteration

$$x_1^{(3)} = \frac{1}{5}(-1 + 2(0.203) - 3(-0.517))$$

$$= \frac{1}{5}(-1 + 0.406 + 1.551) = \frac{1}{5}(0.957) = 0.1914$$

$$x_2^{(3)} = \frac{1}{9}(2 + 3(0.1456) - (-0.517))$$

$$= \frac{1}{9}(2 + 0.4368 + 0.517) = \frac{1}{9}(2.9538) = 0.3282$$

$$x_3^{(3)} = \frac{1}{7}(-3 + 2(0.1456) - 0.203)$$

$$= \frac{1}{7}(-3 + 0.2912 - 0.203) = -\frac{1}{7}(0.2912 - 3.203)$$

$$= -\frac{0.2912}{7} = -0.416$$

$\therefore x^{(3)} = (0.1914, 0.3282, -0.416)$.

For 4th iteration, we will use the values of 3rd iteration

$$x_1^{(4)} = \frac{1}{5}(-1+2(0.3282)-3(-0.416))$$

$$= \frac{1}{5}(-1+0.6564+1.248) = \frac{1}{5}(0.9044) = 0.1808$$

$$x_2^{(4)} = \frac{1}{9}(2+3(0.1914)-(-0.416))$$

$$= \frac{1}{9}(2+0.5742+0.416) = \frac{1}{9}(2.9902) = 0.332$$

$$x_3^{(4)} = \frac{1}{7}(-3+2(0.1914)-0.3282)$$

$$= \frac{1}{7}(-3+0.3828-0.3282) = \frac{1}{7}(0.3828-0.3282)$$

$$= -0.421$$

$$x^4 = (0.1808, 0.3328, -0.421)$$

For 5th iteration,

$$x_1^{(5)} = \frac{1}{5}(-1+2(0.332)-3(-0.421)$$

$$= \frac{1}{5}(-1+0.664+1.263) = \frac{1}{5}(0.927) = 0.1854$$

$$x_2^{(5)} = \frac{1}{9}(2+3(0.1808)+0.421)$$

$$= \frac{1}{9}(2.421+0.5424) = 0.329$$

$$x_3^{(5)} = \frac{1}{7}(-3+2(0.1808)-0.332)$$

$$= \frac{1}{7}(-3.332+0.329, -0.424)$$

$$x^{(5)} = (0.1854\ 0.329, -0.424)$$

For 6th iteration,

$$x_1^{(6)} = \frac{1}{5}(-1 + 2(0.329) - 3(-0.424))$$

$$= \frac{1}{5}(-1 + 0.658 + 1.272)$$

$$= \frac{0.93}{5} = 0.186$$

$$x_2^{(6)} = \frac{1}{9}(2 + 3(0.1854) + 0.424)$$

$$= \frac{1}{9}(2 + 0.5562 + 0.424) = 0.331$$

$$x_3^{(6)} = \frac{1}{7}(-3 + 2(0.1854) - 0.329)$$

$$= \frac{1}{7}(-3 + 0.3708 - 0.329) = \frac{-2.9582}{7} = -0.422$$

$$x^{(6)} = (0.186, 0.331, -0.422)$$

In 5th iteration and 6th iteration, two successive approximation are identical.

Ex:- Apply Gauss-seidal method to solve,

$$5x_1 - 2x_2 + 3x_3 = -1$$
$$-3x_1 + 9x_2 + x_3 = 2$$
$$2x_1 - x_2 - 7x_3 = 3$$

$$\therefore x_1 = \frac{1}{5}(-1 + 2x_2 - 3x_3)$$
$$x_2 = \frac{1}{9}(2 + 3x_1 - x_3)$$
$$x_3 = \frac{1}{7}(-3 + 2x_1 - x_2) \quad (*)$$

Consider the initial approximation $x^{(0)} = (0,0,0)$

$$x_1^{(1)} = \frac{1}{5}(-1 + 0 - 0) = -0.2$$

$$x_2^{(1)} = \frac{1}{9}(2 + 3(-0.2) - 0) = \frac{1}{9}(2 - 0.6) = 0.155$$

$$x_3^{(1)} = \frac{1}{7}(-3 + 2(-0.2) - 0.155)$$

$$= \frac{1}{7}(-3 - 0.4 - 0.155) = \frac{-3.555}{7} = -0.507$$

$$x^{(1)} = (-0.2, 0.155, -0.507)$$

For 2nd iteration, we will consider values of $x^{(1)}$

$$x_1^{(2)} = \frac{1}{5}(-1 + 2(0.155) - 3(-0.507))$$

$$= \frac{1}{5}(-1 + 0.31 + 1.521) = \frac{0.831}{5} = 0.166$$

$$x_2^{(2)} = \frac{1}{9}(2 + 3(0.166) + 0.507)$$

$$= \frac{1}{9}(2 + 0.498 + 0.507) = \frac{3.0056}{9} = 0.334$$

$$x_3^{(2)} = \frac{1}{7}(-3 + 2 \times 0.166 - 0.334) = \frac{1}{7}(-3 + 0.332 - 0.334)$$

$$= \frac{1}{7}(-3.002) = -0.428$$

$$x^{(2)} = (0.166, 0.334, -0.428)$$

We start 3rd iteration,

$x_1^{(3)} = \frac{1}{5}(-1+2(0.334) - 3(-0.428))$

$= \frac{1}{5}(-1+0.668+1.284) = \frac{0.952}{5} = 0.1904$

$x_2^{(3)} = \frac{1}{9}(2+3(0.1904)+0.428)$

$= \frac{1}{9}(2+0.5712+0.428) = 0.333$

$x_3^{(3)} = \frac{1}{7}(-3+2(0.1904)-0.333)$

$= \frac{1}{7}(-3+0.3808-0.333) = \frac{1}{7}(0.3808-3.333)$

$x^{(3)} = (0.1904, 0.333, -0.421) \quad = \frac{-2.9522}{7} = -0.421$

We start 4th iteration,

$x_1^{(4)} = \frac{1}{5}(-1+2(0.333)-3(-0.421)$

$= \frac{1}{5}(-1+0.666+1.263) = 0.1858$

$x_2^{(4)} = \frac{1}{9}(2+3(0.1858)+0.421)$

$= \frac{1}{9}(2+0.5574+0.421) = 0.3309$

$x_3^{(4)} = \frac{1}{7}(-3+2(0.1858)-0.3309)$

$= \frac{1}{7}(-3+0.3716-0.3309)$

$= \frac{1}{7}(0.3716-3.3309)$

$= -0.422$

$x^{(1)} = (0.1858, 0.33091, -0.422)$

We start 5th iteration,

$x_1^{(5)} = \frac{1}{5}(-1+2(0.3309)-3(-0.422))$

$= \frac{1}{5}(-1+0.6618-1.266) = 0.1855$

$x_2^{(5)} = \frac{1}{9}(2+3(0.1855)+0.422)$

$= \frac{1}{9}(2+0.5566+0.422) = 0.3309$

$x_3^{(5)} = \frac{1}{7}(-3+2(0.1855)-0.3309)$

$= \frac{1}{7}(-3+0.371-0.3309)$

$= \frac{1}{7}(0.371-3.3309)$

$= -0.422$

$x^{(5)} = (0.1855, 0.3309, -0.422)$

H.W

* $26x_1 + 2x_2 + 2x_3 = 12.6$
  $3x_1 + 27x_2 + x_3 = -14.3$
  $2x_1 + 3x_2 + 17x_3 = 6$

Jacobi method and Gauss-seidal meth

* $45x_1 + 2x_2 + 3x_3 = 58$
  $-3x_1 + 22x_2 + 2x_3 = 47$
  $5x_1 + x_2 + 20x_3 = 67$

~~Solution~~ Answer slide 10 ও আছে

H.W assignment

11/11/2026 — State CHT theorme (একাদশ আধারে) 1 মার্ক প্রশ্নমালায়

Caylay-Hamilton theorem:-

Every matrix A is a root of its characteristic polynomial.

*Verify CHT for $A = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$

Sol^n:-

Characteristic matrix of A is

$$\lambda I - A = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \lambda-1 & -2 \\ 1 & \lambda-1 \end{pmatrix}$$

Characteristic polynomial of A is $|\lambda I - A|$

$$= \begin{vmatrix} \lambda-1 & -2 \\ 1 & \lambda-1 \end{vmatrix}$$

$$= (\lambda-1)^2 + 2 = \lambda^2 - 2\lambda + 3$$

The characteristic ~~eqn~~ equation of A is $\lambda^2 - 2\lambda + 3 = 0$. To verify CHT we have to show that $A^2 - 2A + 3I = 0$

Here $A^2 - 2A + 3I$

$$= \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} - 2\begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} + 3\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 4 \\ -2 & -1 \end{pmatrix} - \begin{pmatrix} 2 & 4 \\ -2 & 2 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Here A verifies the CHT

প্রশ্নে অন্যরকম থাকলে প্রথমে (আগেই) Change করবে

* If $A=\begin{pmatrix}1 & 1\\ 1 & 2\end{pmatrix}$, find A^{-1}, A^{-2}, A^{-3} using CHT

Soln:-

Characteristic polynomial of A is

$$|\lambda I - A| = \begin{vmatrix}\lambda-1 & -1\\ -1 & \lambda-2\end{vmatrix} = (\lambda-1)(\lambda-2)-1 = \lambda^2-3\lambda+1$$

Characteristic equation is $\lambda^2-3\lambda+1=0$.

Thus A satisfies the equation $A^2-3A+I=0$ ..... (1)

Multiplying (1) by A^{-1} we get,

$$A-3I+A^{-1}=0$$

$$\Rightarrow A^{-1}=3I-A \quad \ldots\ldots (2)$$

$$A^{-1}=3\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}-\begin{pmatrix}1 & 1\\ 1 & 2\end{pmatrix}=\begin{pmatrix}2 & -1\\ -1 & 1\end{pmatrix}$$

Multiplying (2) by A^{-1}

$$A^{-2}=3A^{-1}-I \quad \ldots\ldots (3)$$

$$A=3\begin{pmatrix}2 & -1\\ -1 & 1\end{pmatrix}-\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=\begin{pmatrix}5 & -3\\ -3 & 2\end{pmatrix}$$

Again multiplying (3) by A^{-1}, we get,

$$A^{-3}=3A^{-2}-A^{-1}$$

$$=3\begin{pmatrix}5 & -3\\ -3 & 2\end{pmatrix}-\begin{pmatrix}2 & -1\\ -1 & 1\end{pmatrix}$$

$$=\begin{pmatrix}15 & -9\\ -9 & 6\end{pmatrix}-\begin{pmatrix}2 & -1\\ -1 & 1\end{pmatrix}$$

$$=\begin{pmatrix}13 & -8\\ -8 & 5\end{pmatrix}$$

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find K (3 by 3) matrix

*1 a matrix P that diagonalizes $A\begin{bmatrix}6 & 0\\ -1 & \end{bmatrix}$ and also determine $P^{-1}AP$

① Soln:- Characteristic polynomial of A is

$$|\lambda I - A| = \begin{vmatrix}\lambda-1 & 0\\ -6 & \lambda+1\end{vmatrix} = (\lambda-1)(\lambda+1)+0 = (\lambda+1)(\lambda-1)$$

So characteristic equation of A is $(\lambda-1)(\lambda+1)=0$

$\therefore \lambda = 1, -1$, which are the eigen values of A

By defn of eigen vector of a matrix, $\underline{v} = \begin{pmatrix}x_1\\ x_2\end{pmatrix}$ is a non-zero eigen vector of A corresponding to eigenvalue 1)

if $(A-\lambda I)\underline{v} = 0$

$$\therefore \begin{bmatrix}\lambda-1 & 0\\ -6 & \lambda+1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\Rightarrow \begin{cases}(\lambda-1)x_1 = 0\\ -6x_1 + (\lambda+1)x_2 = 0\end{cases} \quad \cdots\cdots (1)$$

When $\lambda = 1$, we get from (1)

$$-6x_1 + 2x_2 = 0$$

$$\Rightarrow 3x_1 - x_2 = 0$$

The system has a non-zero solution. Here x_2 is a free variable.

$$x_2 = 3,\ x_1 = 1 \qquad \underline{v_1} = \begin{bmatrix}1\\ 3\end{bmatrix}$$

When $\lambda = -1$, we get from (1)

$$\begin{cases}-2x_1 = 0 \Rightarrow x_1 = 0\\ -6x_1 = 0\end{cases}$$

This system has non-zero solution. Here x_2 is a free variable at $x_2 = 1$ $\therefore \underline{v_2} = \begin{bmatrix}0\\ 1\end{bmatrix}$

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Let P be the matrix whose columns are the above two vectors

$$\therefore P = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$$|P| = 1 \neq 0$$

$$P^{-1} = \frac{1}{1}\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

Now,

$$P^{-1}AP = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Which is the diagonal matrix of eigenvalues of A.

2. Find the matrix P that diagonalies $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$

Sol^n:- The characteristic polynomial of A is

$$|\lambda I - A| = \begin{vmatrix} \lambda & 0 & 2 \\ -1 & \lambda-2 & -1 \\ -1 & 0 & \lambda-3 \end{vmatrix}$$

$$= \lambda\{(\lambda-2)(\lambda-3)-0\} + 0 + 2\{0+(\lambda-2)\}$$
$$= \lambda(\lambda^2-5\lambda+6) + 2\lambda - 4$$
$$= \lambda^3 - 5\lambda^2 + 8\lambda - 4$$
$$= \lambda^2(\lambda-1) - 4\lambda(\lambda-1) + 4(\lambda-1)$$
$$= (\lambda-1)(\lambda-2)^2$$

So characteristic equation of A is $(\lambda-1)(\lambda-2)^2 = 0$

which are the eigen values of A $\therefore \lambda = 1, 2, 2$

Now, to find the eigen vector $\underline{v} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ corresponding to λ, solve the homogeneous linear system ④ represted by $(\lambda I - A)\underline{v} = 0$

$$\therefore \begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda-2 & -1 \\ -1 & 0 & \lambda-3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \lambda x_1 + 2x_3 = 0 \\ -x_1 + (\lambda-2)x_2 - x_3 = 0 \\ -x_1 + (\lambda-3)x_3 = 0 \end{cases} \quad \text{—— ①}$$

When $\lambda = 2$, we get from ①

$$\begin{cases} 2x_1 + 2x_3 = 0 \\ -x_1 - x_3 = 0 \\ -x_1 - x_3 = 0 \end{cases} \Rightarrow x_1 + x_3 = 0$$

The system has non-zero solution. Here x_2 and x_3 are free variables let $x_2 = 5$ and $x_3 = t$ where $s, t \in \mathbb{R}$. then $x_1 = -t$

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Thus solution of the above ~~equs~~ system is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{pmatrix} -t \\ s \\ t \end{pmatrix} = s\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Therefore the non-zero eigen vectors of A corresponding to the eigenvalue $\lambda = 2$ are

$$\underline{v} = \begin{bmatrix} -t \\ s \\ t \end{bmatrix} = s\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Thus the non-zero eigen vectors at A corresponding to the eigen value $\lambda = 2$ and

$\underline{v_1} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$; $\underline{v_2} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

When $\lambda = 1$, we get from (i)

$$\begin{cases} x_1 + 2x_3 = 0 \\ -x_1 - x_2 - x_3 = 0 \\ -x_1 - 2x_3 = 0 \end{cases}$$

$$\sim \begin{cases} x_1 + x_2 + x_3 = 0 \\ x_1 + 2x_3 = 0 \end{cases}$$

$$\sim \begin{cases} x_1 + x_2 + x_3 = 0 \\ -x_2 + x_3 = 0 \end{cases}$$

The system has a non-zero solution. Here x_3 is a free variable. at $x_3 = 1$, thus $x_2 = 1$, $x_1 = -2$

Thus the non-zero eigen vector of A corresponding to the eigen value $\lambda = 1$ is

$\underline{v_3} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$

Therefore, the three basis vectors are

$$\underline{v_1} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \underline{v_2} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad \underline{v_3} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

Let P be the matrix where colums are the above three vectors, which diagonlizes A

$$\therefore P = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$|P| = -1 \neq 0$$

Cofactors of P are $P_{11} = -1 \quad P_{12} = -1 \quad P_{13} = 1$

$P_{21} = -1 \quad P_{22} = 0 \quad P_{23} = 0$

$P_{31} = -1 \quad P_{32} = -2 \quad P_{33} = 1$

$$\therefore P^{-1} = \frac{1}{-1}\begin{pmatrix} -1 & -1 & -1 \\ -1 & 0 & -2 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ -1 & 0 & -1 \end{pmatrix}$$

$$\text{Now, } P^{-1}AP = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ -1 & 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}\begin{pmatrix} 0 & -1 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ -1 & 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & -2 & -2 \\ 2 & 0 & 1 \\ 0 & 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

which is a diagonal matrix.

Thus the matrix P diagonalizes A

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